UM2Former: U-Shaped Multimixed Transformer Network for Large-Scale Hyperspectral Image Semantic Segmentation

Aijun Xu, Zhaohui Xue, *Member, IEEE*, Ziyu Li, Shun Cheng, Hongjun Su, *Senior Member, IEEE*, and Junshi Xia, *Senior Member, IEEE*

***Abstract*— Transformer-based deep learning (DL) methods have gradually been advocated for remote sensing (RS) image semantic segmentation due to the great global modeling capability. Nevertheless, Transformer-based DL methods have not yet been sufficiently explored on the large-scale hyperspectral image (HSI) semantic segmentation. Current algorithms lack a comprehensive consideration of the impact of positional encoding (PE) interpolation when constructing Transformer-based decoders. Moreover, existing segmentation heads usually directly concatenate multiscale features to achieve segmentation, which ignores the inherent semantic differences between different features. To address the above issues, a U-shaped multimixed Transformer network (UM2Former) is proposed for large-scale HSI semantic segmentation. First, a weight encoder consisting of two modules, the overlap-down and the channel-weight, is built to extract hierarchical discriminative spectral–spatial features and decrease spectral redundancy. Second, the proposed multimixed Transformer block (MMTB) develops a PE-free module, spatial-feature-retention attention (SFRA) mechanism, in which "multimixed" represents the global dependency modeling of each pixel with the retented average spatial characteristics of different locations in the input feature maps. Finally, a linear fuse segmentation head (LFSH) is designed to align semantic information among multiscale feature maps and achieve accurate segmentation. Experiments were conducted in single cities and the entire large-scale WHU-OHS HSI dataset. The segmentation results indicated that the proposed method achieved higher accuracy compared to the existing semantic segmentation methods, with performance improvements of 17.80% and 4.16% in terms of intersection over union (mIoU) and overall accuracy (OA), respectively. The source code will be available at https://github.com/ZhaohuiXue/UM2Former.**

***Index Terms*— Hierarchical structure, large-scale hyperspectral image (HSI), positional encoding (PE), semantic segmentation, Transformer.**

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Aijun Xu, Zhaohui Xue, Ziyu Li, Shun Cheng, and Hongjun Su are with the College of Geography and Remote Sensing, Hohai University, Nanjing 211100, China (e-mail: zhaohui.xue@hhu.edu.cn).

Junshi Xia is with RIKEN Center for Advanced Intelligence Project (AIP), Tokyo 103-0027, Japan.

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I. Introduction

THE acquisition of massive amounts of hyperspectral image (HSI) due to the rapid advancement of aerospace technology and sensor technology poses great challenges for efficient processing and analysis of large-scale HIS [1]. Semantic segmentation, as a feasible solution [2], is to learn discriminative information from a vast number of annotated samples and identify the semantic category of each pixel in the image. Currently, semantic segmentation has been widely applied in many remote sensing (RS) image segmentation tasks, including land cover mapping [3], [4], [5], water resources management [6], [7], and road and building extraction [8], [9], [10].

In the past decade, convolutional neural networks (CNNs) have been widely and successfully applied in RS semantic segmentation [11], [12], [13]. CNNs possess a strong inductive bias, characterized by spatial locality and translation invariance, which leads to excellent refinement of local features [14]. However, these also make CNNs unable to mine the most important features and lack the capability to ignore irrelevant information [15], [16], [17]. Attention mechanisms compensate for these deficiencies, which can extract features strongly related to the target task and weaken the weight of noisy information [18], and has been widely studied to enhance the performance of semantic segmentation models [19]. Generally, attention mechanisms can be directly embedded into CNN-based deep learning (DL) frameworks as a plug-and-play module [20], [21], [22]. However, an over-reliance on convolutional operation to process features derived from attention mechanisms will limit the representation of global features. In recent years, Transformer [23] emerged as a completely different approach from CNN in image processing tasks. Transformer has a strong ability to model the global dependencies, particularly demonstrated by vision transformer (ViT) [24] and its variants outperforming CNN-based models in image classification and semantic segmentation [25], [26], [27]. For example, Wang et al. [25] proposed a pyramid vision transformer (PVT) to learn multiscale and high-resolution features by stacking multiple individual Transformer encoders. And then, Liu et al. [26] designed a hierarchical SwinTransformer with window multihead self-attention (W-MSA) module

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to decrease the quantity of computation and a shifted windows multihead self-attention (SW-MSA) module to bridge information between different windows, and handle the correlation between them. To tackle the problem of semantic ambiguity between pixels in RS scenarios, Bi et al. [27] proposed an Agent Mining Transformer (AgMT) to adaptively mine a set of local-aware agents and construct agent-level semantic correlation which contains the global–local information. Transformers with hierarchical structure have shown excellent performance in natural scene image processing tasks and have been widely adopted in RS semantic segmentation [28], [29], [30].

Although CNN-based and Transformer-based models have demonstrated impressive results in RS semantic segmentation [31], these models may not be appropriate for analyzing complex HSI due to inherent subpixel and high spectral resolution characteristics [32]. The reason is that the accuracy of RS semantic segmentation heavily relies on the effectiveness of high-resolution feature extraction networks, whereas the accuracy of HSI semantic segmentation is significantly influenced by both subpixel and spectral variability.

In order to enable DL-based models to extract complex HSI features effectively, a lot of previous works are patch-based learning strategies [33], [34], [35]. The CNN-based methods aim to explore spectral–spatial features from adjacent pixels within a square box in an HSI patch for HSI classification. According to different convolutional kernel operation rules, CNN can be divided into 1-D CNN [36], 2-D CNN [37], 3-D CNN [33], and multichannel CNN [38]. However, the overlapping image patches generated by adjacent pixels cause computational redundancy [39], slowing down the patch-based learning framework. Additionally, capturing only local spatial features within the patch fails to obtain long-range spatial dependencies. Recently, a patch-free learning framework uses the whole image as input to achieve end-to-end classification, which is similar to semantic segmentation. To address the above issues, Xu et al. [40] first proposed a spectral–spatial fully convolutional network (SSFCN) to effectively avoid the limitations inherent in patch classification methods. SSFCN employs a double-branch architecture to separately extract spectral and spatial features, and the dense conditional random field (CRF) is integrated into the network to further balance the local and global information. Subsequently, Zheng et al. [39] proposed a fast patch-free global learning (FPGA) framework for HSI semantic segmentation, which extracts global spatial contextual information for the whole image via a spectral attention-based encoder and a lightweight decoder structure, and obtains diverse gradients from stratified samples to guarantee the convergence. To distinguish similar land covers effectively, Zhu et al. [41] proposed a global convolutional long short-term memory (GCL) module, and designed a global joint attention mechanism (GJAM) to reweight and model the extracted features. In addition, Yu et al. [42] proposed a channel convolutional long short-term mechanism and multiorder spatial aggregation block (CLMS) to extract more discriminative spectral–spatial features, and designed a pyramid-enhanced attention mechanism (PEAM) in the decoder to retain more spatial information. Very recently, a content-driven spectrum complementary network (CSCN) was proposed in [43], where content-adaptive pointwise fusion module (CPFM) in the decoder is used to aggregate and refine features from the dual encoder along with the fused features from the previous stage through self-attention mechanism.

Although HSI classification and semantic segmentation methods based on FCN, and attention mechanism have demonstrated promising results, several important issues remain to consider. How does the decoder retain more spatial information and effectively address the scale diversity of various land cover types? How to take advantage of multiscale feature maps in context more efficiently? In addition, the abovementioned methods are all experimented on single-scene hyperspectral datasets, and few efforts have been made for algorithm tests and applications on large-scale HSI due to the scarcity of publicly available datasets [44]. Recently, Li et al. [45] built a dense and large HSI benchmark dataset: WHU-OHS, the large spatial coverage and extensive classification scene make the WHU-OHS dataset a challenging benchmark for large-scale HSI semantic segmentation.

The abovementioned semantic segmentation methods have achieved excellent performance, and there are still several limitations.

1) Natural and RS images are often characterized by high spatial resolution, while HSIs are distinguished by their high spectral resolution and pervasive subpixel. The accuracy of RS semantic segmentation heavily relies on the effectiveness of high-resolution feature extraction networks, whereas the accuracy of HSI semantic segmentation is significantly influenced by both subpixel and spectral variability. Consequently, the feature extraction network of RS semantic segmentation needs to be redesigned to extract more discriminative spectral–spatial features of HSI [46].
2) To model the global dependencies, mainstream solutions focus on employing Transformer blocks as the encoder. This design loses spatial information and requires positional encoding (PE) to retain part of spatial location information. In contrast, when the Transformer blocks are introduced as a decoder to process feature maps with abundant spatial information, the interpolation of PE will lead to decreased performance.
3) In the upsampling process of the hierarchical decoder, the high-level semantic features output by different layers contain different granularity information. Nevertheless, there are limited methods for fusing multiscale features while considering the inherent semantic differences between different features.

Very recently, a simple and efficient semantic segmentation network called SegFormer is introduced [47], which considers the effect of zero padding on generating location information and proposes to replace the PE in Transformer with the combination of an overlapped patch merging (a 3×3 convolutional layer with a stride of 2) and a 3×3 convolution in the feed-forward network. SegFromer proves that PE is not necessary for semantic segmentation. Furthermore, U-NetFormer [48] innovates to design a

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Transformer module to construct the decoder instead of the encoder utilized to extract the feature. The lateral connection leverages the spatial detail features of the shallow convolutional layers and passes more precise locations of features from the encoder to the decoder. We argue that this design could overcome the limitations imposed by PE, although U-NetFormer retains the PE.

In this article, we are inspired by SegFormer and U-NetFormer and propose a U-shaped multimixed Transformer network (UM2Former) for large-scale HSI semantic segmentation. Specifically, UM2Former builds a weight encoder to extract hierarchical discriminative spectral–spatial features and decrease the spectral redundancy. To avoid the limitations that may arise from the interpolation of PE and the loss of spatial information, we present a multimixed Transformer block (MMTB) to construct the hierarchical Transformer decoder. Unlike the Transformer-based encoder for feature extraction, the MMTB is specifically designed to process feature maps with precise spatial information. To achieve this, the MMTB develops a spatial-feature-retention attention (SFRA) mechanism to retent the spatial average characteristics of different locations and then model the global dependency of each pixel in the input feature maps with these characteristics. Finally, considering the inherent semantic differences between different features, a lightweight linear-fuse segmentation head (LFSH) composed of multiple linear layers and a semantic fusion module is designed to fuse the multiscale features adaptively.

The main innovative contributions of this article can be summarized as follows.

1) Consider the intrinsic property of spectral–spatial unification in HIS. Unlike traditional semantic segmentation backbones, our weight encoder emphasizes the spectral–spatial information extraction capabilities of the feature extraction module. By combining overlap-down and channel-weight modules, the weight encoder initially extracts continuous local fine-grained feature maps and subsequently reweights these maps based on global context to select more discriminative spectral features.
2) UM2Former revisits the structure of the encoder and decoder in semantic segmentation models and designs MMTB to construct Transformer-based decoder. MMTB introduces a PE-free module, the SFRA layer, to model the global dependency of each pixel in the weighted feature map with the spatial retention feature. This module replaces the multihead self-attention (MHSA) module in traditional Transformer models and greatly reduces the number of parameters.
3) Considering the inherent semantic differences between output feature maps of the decoder at different stages, the proposed LFSH is designed to progressively map the features output by the decoder through linear hierarchical alignment. By utilizing learnable parameters, LFSH optimizes the semantic consistency between shallow detail features and deep features. This approach effectively reduces the semantic gap across different features, refining the final semantic mask.

II. Proposed Methodology

As illustrated in Fig. 1, the proposed UM2Former comprises three main modules. In the first module, the weight encoder block is designed to extract hierarchical discriminative spectral–spatial feature maps over the entire input HSI and then decrease spectral redundancy. Then, the semantic features produced by the weight encoder are aggregated with the features generated by the MMTB of the decoder using a weighted fusion (WF) operation. Second, MMTB models the dependency of each pixel in the efficient feature map with spatial average characteristics of different locations. Finally, LFSH is attached at the end of the UM2Former to adaptively fuse the multiscale features and then generate the final segmentation mask. Module details and the input and output sizes of each module in the standard UM2Former are presented in Table I.

A. Overview Architecture

Fig. 1 illustrates our proposed UM2Former, a multistage feature extraction architecture with four hierarchical stages. In each stage, local fine-grained features with discriminative channel information are first extracted using a weight encoder, and then a WF operation is followed to learn more generalized fusion features. MMTB is performed after each WF operation to mine the global dependency of each pixel in the weighted feature map with the spatial retention feature, which acts as one input for the next stage of WF. This design enables global correlation extraction from shallow detail features to deep semantic features, fully exploiting the complementary cues underlying different stages of features. For instance, the fusion of shallow detail features helps to mitigate the loss of spatial details that occurs during the downsampling process. This is particularly significant in the context of HSI, which often suffers from limited spatial resolution. Finally, the features that have undergone multistage learning are fed into LFSH to align semantic information and achieve accurate segmentation.

B. Weight Encoder

The encoder in UM2Former follows a modular design, made up of four hybrid blocks, all of which contain the overlap-down and the channel-weight modules. Overlap-down module is a 4×4 convolutional layer with a stride of 2 followed by group normalization (GN) [49] and rectified linear unit (ReLU) activation. Compared with the commonly used downsampling module like 3×3 maxpool operation with a stride of 2 or 3×3 dilated convolution [50] with a dilation ratio of 2, overlap-down module enables the extraction of continuous local detail features while preserving the actual receptive field, thereby providing more effective and restorable spatial information for the decoder. The channel-weight module reweights the feature map via global context and highlights the more important feature map to improve the accuracy of the model, which is a similar implementation to the SE-Block [51].

As shown in Fig. 2, in any layer of the proposed encoder, given the original HSI data $X = (X_1, X_2, \cdots, X_C) \in \mathbb{R}^{C \times H \times W}$, H, W, and C denote the height, width, and

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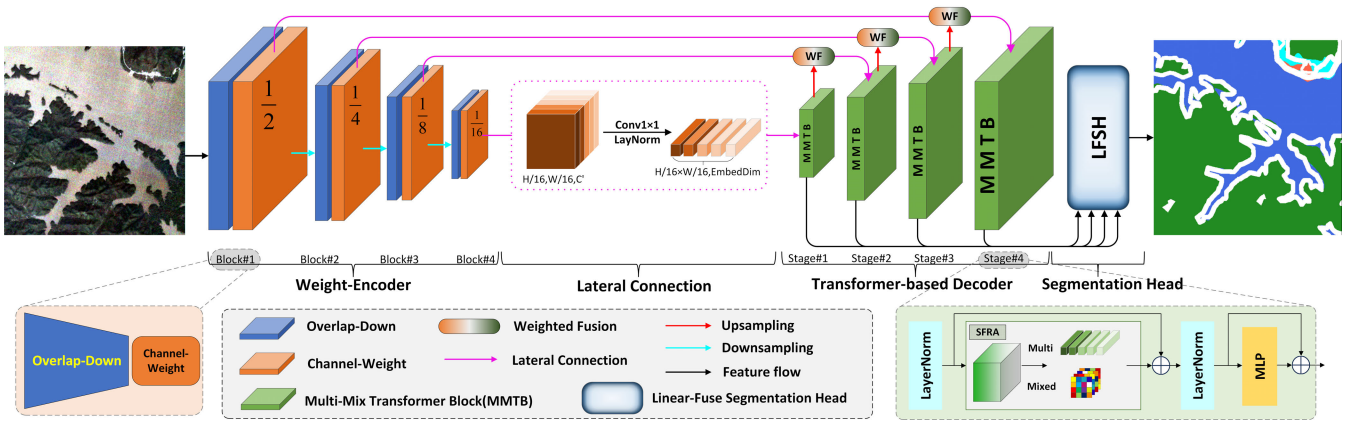


Fig. 1. Graphical illustration of the proposed UM2Former.

TABLE I

ARCHITECTURE DETAILS OF THE STANDARD UM2FORMER (UM2FORMER WITH OP = 4)

Structure	Section	Module	Input size	Specification	Output size
Input layer	-	-	$(512, 512, 32)$	-	-
Encoder	Block#i $(i = 1, 2, 3, 4)$	Overlap-Down	$\left(\frac{512}{2^{i-1}}, \frac{512}{2^{i-1}}, 2^{i+4}\right)$	4×4Conv, S = 2	$\left(\frac{512}{2^i}, \frac{512}{2^i}, 2^{i+5}\right)$
		Channel-Weight	$\left(\frac{512}{2^i}, \frac{512}{2^i}, 2^{i+5}\right)$	Attention (r=16)	$\left(\frac{512}{2^i}, \frac{512}{2^i}, 2^{i+5}\right)$
			$\left(\frac{512}{2^i}, \frac{512}{2^i}, 2^{i+5}\right)$	3×3Conv, S = 2	$\left(\frac{512}{2^i}, \frac{512}{2^i}, 2^{i+5}\right)$
Lateral	Lateral#1	Connection	$(32, 32, 512)$	1×1Conv$(dim = 128)$, S = 1	$(128, 32^2)$
	Lateral#j $(j = 2, 3, 4)$	Connection	$(2^{j+4}, 2^{j+4}, 128)$	1×1Conv$(dim = 128)$, S = 1	$(128, 2^{(j+4)\times 2})$
		Weighted Fusion	$(128, 2^{(j+3)\times 2})$	Upsampling×2	$(128, 2^{(j+4)\times 2})$
			$(128, 2^{(j+4)\times 2})$	Learnable parameters: α, $(1-\alpha)$	$(128, 2^{(j+4)\times 2})$
Decoder	MMTB Stage#k $(k = 1, 2, 3, 4)$	LayerNorm	$(128, 2^{(k+4)\times 2})$	LayerNorm$(dim = 128)$	$(128, 2^{(k+4)\times 2})$
		SFRA	$(128, 2^{(k+4)\times 2})$	SFRA$(head = 8, SFR_{ratio} = 2^{k+1})$	$(128, 2^{(k+4)\times 2})$
		LayerNorm	$(128, 2^{(k+4)\times 2})$	LayerNorm$(dim = 128)$	$(128, 2^{(k+4)\times 2})$
		MLP	$(128, 2^{(k+4)\times 2})$	MLP$(MLP_{ratio} = 2)$	$(128, 2^{(k+4)\times 2})$
Segmentation Head	LFSH	Multi Linear	$(128, 2^{(i+4)\times 2})$	4×Linear$(dim = 128)$	$(128, 2^{(i+4)\times 2})$
		Semantic Fusion	$(128, 32^2)$	Upsampling×8	$(128, 256^2)$
			$(128, 64^2)$	Upsampling×4	$(128, 256^2)$
			$(128, 128^2)$	Upsampling×2	$(128, 256^2)$
			$4\times(128, 256^2)$	Concat$(dim = 4 \times 128)$	$(4 \times 128, 256^2)$
			$(4 \times 128, 256^2)$	Linear$(dim = 128)$	$(128, 256^2)$
		ConvClass	$(128, 256, 256)$	3×3Conv$(dim = C)$, S = 2	$(C, 256, 256)$
			$(C, 256, 256)$	1×1Conv$(dim = N_{Class})$, S = 1	$(N_{Class}, 256, 256)$
			$(N_{Class}, 256, 256)$	Upsampling×2	$(N_{Class}, 512, 512)$

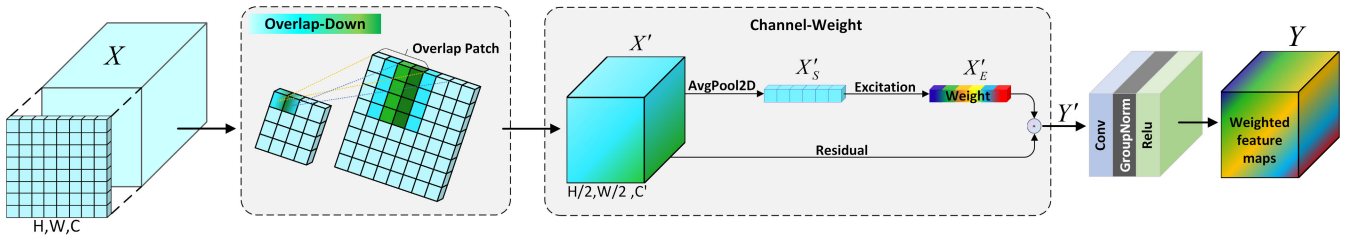


Fig. 2. Graphical illustration of the proposed weight encoder block.

number of spectral bands, respectively. We first compute the downsampling tensor $X' \in \mathbb{R}^{C'\times(H/2)\times(W/2)}$, C' denotes the dimension of the feature maps after downsampling

$$X' = \delta\left(\mathrm{GN}\left(X'_1, \cdots, X'_i, \cdots, X'_{C'}\right)\right) \tag{1}$$

$$X'_i = V_i * X = \sum_{m=1}^{C} V_i^m * X_m \tag{2}$$

where $V = [V_1, \cdots, V_i, \cdots, V_{C'}]$ is a set of learnable filter kernels, when $V_i = [V_i^1, V_i^2, \cdots, V_i^C]$ acts on the input tensor X. Here $*$ denotes convolution, δ denotes ReLU, and GN indicates the group normalization function.

In order to highlight the more important feature map and suppress less useful ones, channel-weight module models the channel dependencies via global context information in $X'_1, X'_2, \cdots, X'_{C'}$. First, we compute the global context information description vector $X'_S \in \mathbb{R}^{C'\times 1\times 1}$

$$X'_S(C', 1, 1) = \frac{4}{H \cdot W} \sum_{i=1}^{\frac{H}{2}} \sum_{j=1}^{\frac{W}{2}} X'(C', i, j). \tag{3}$$

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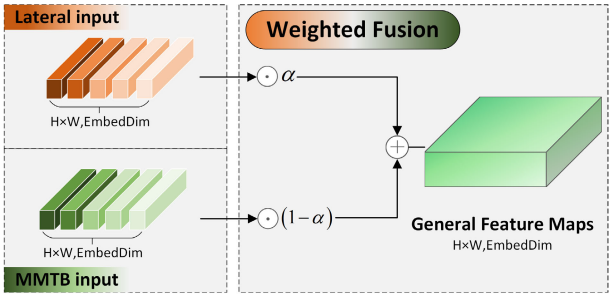


Fig. 3. Graphical illustration of the WF.

And then, in order to learn a nonlinear interaction between channels flexibly and achieve multichannel activation, we use two fully connected (FC) layers followed by a sigmoid gating mechanism to compute the channel weight vector $X'_E \in \mathbb{R}^{C'\times 1\times 1}$

$$X'_E\left(C', 1, 1\right) = \text{sigmoid}\left(W_2 \times \delta\left(W_1 \times X'_S\right)\right) \tag{4}$$

where $W_1 \in \mathbb{R}^{(C'/r)\times C'}$, $W_2 \in \mathbb{R}^{C'\times(C'/r)}$, and sigmoid denotes the sigmoid mechanism. To reduce the parameters and model complexity, the reduction ratio r is set to 16, and this parameter setting is discussed in Section III-C3. The weighted feature map $Y \in \mathbb{R}^{C'\times(H/2)\times(W/2)}$ is the output of weight encoder

$$Y' = X'_E\left(C', 1, 1\right) \cdot X'\left(C', \frac{H}{2}, \frac{W}{2}\right) \tag{5}$$

$$Y = \delta\left(\text{GN}\left(Y'\right)\right). \tag{6}$$

Considering that the weighted feature map generated by each encoder stage contains suppressed parts, directly fusing it with the corresponding decoder feature map may weaken the feature representation of the decoder. Our lateral connection is implemented by a 1×1 convolution with channel dimension in 128, which preserves the most effective feature map and provides more precise locations of features for the decoder. And then, the semantic features produced by the weight encoder are aggregated with the features generated by the MMTB of the decoder using a WF operation. As shown in Fig. 3, to learn more generalized fused features, the WF operation weights two features selectively based on their contributions to segmentation accuracy [48], [52]. The formulation can be denoted as follows:

$$\hat{\text{WF}}_i = \alpha \cdot \text{Linear}(WE_{4-i}) + (1-\alpha)\cdot(\text{MMF}_i), i = 1, 2, 3 \tag{7}$$

where $\hat{\text{WF}}_i \in \mathbb{R}^{128\times HW}$ represents the general feature map, $WE_{4-i} \in \mathbb{R}^{128\times HW}$ denotes the efficient feature map produced by the 4-i layer weight encoder, MMF $\in \mathbb{R}^{128\times(HW/4)}$ indicates the feature generated by the i stage MMTB.

C. Multimixed Transformer Block

The MMTB consists of the SFRA, multilayer perceptron (MLP), layer normalization layers, and residual connection operation.

Inspired by efficient self-attention in SegFormer and global–local attention in U-NetFormer, we design an MMTB to construct the hierarchical decoder. Unlike the Transformer-based encoder for feature extraction, the MMTB needs to process feature maps with precise spatial information, we propose an SFRA layer to replace the MHSA layer and avoid the interpolation of PE. Technically, we first perform layer normalization operation on the feature maps generated by WF. Then, we apply a linear layer to obtain the query (Q) vectors and apply an SFR layer to compute the key (K), value (V) matrix

$$\text{WF}_i = \text{LayerNorm}\left(\hat{\text{WF}}_i\right) \tag{8}$$

$$Q = \text{Reshape}(\text{Linear}(\text{WF}_i)) \tag{9}$$

$$K = V = \text{Reshape}(\text{SFR}(\text{WF}_i)) \tag{10}$$

where $\text{WF}_i \in \mathbb{R}^{128\times HW}$, and $Q \in \mathbb{R}^{(128/h_i)\times HW\times h_i}$ is essentially all the pixels in the feature maps, representing the "Multi" tokens in MMTB. h_i is the head number of the SFRA layer in MMTB stage#i. Therefore, the dimension of each head is equal to $(128/h_i)$. r_i is the retent ratio of the SFRA layer in MMTB stage#i. In our experiments, we set r_i to [4, 8, 16, 32] from MMTB stage#1 to stage#4. SFR$(\cdot)$ is the operation for retenting the spatial average characteristics of different locations, so $K, V \in \mathbb{R}^{(128/h_i)\times(H/r_i)\times(W/r_i)\times h_i}$ is essentially spatial retention feature of feature maps, representing the "Mixed" tokens in MMTB

$$\text{SFR}(\text{WF}_i) = \text{Linear}(\text{AvgPool}(\text{Reshape}(\text{WF}_i), r_i)) \tag{11}$$

where Reshape(WF_i) is an operation of reshaping the input sequence WF_i to a feature map $\hat{X}$ of size $128 \times H \times W$. Avgpool$(\hat{X}, r_i)$ is designed to retent the spatial average characteristics of $\hat{X}$. In the original MHSA, each of the heads Q, K, V have the same dimensions $HW \times C$, our attention layer is estimated as follows:

$$\text{SFRA}(Q, K, V) = \text{Linaear}\left(\text{Softmax}\left(\frac{QK^T}{\sqrt{\frac{128}{h_i}}}\right)V\right) \tag{12}$$

The essence of QK^T is to model the global dependence between each pixel in the feature maps and spatial average characteristics of different locations. This is the key to reduce the computational overhead and improve the efficiency of MMTB.

The computational complexity of MSHA is $O((HW)^2)$, ignore the computational costs of the linear layer, the computational complexity of our attention operation is $O(((HW)^2/r_i^2))$, which means our SFRA can effectively reduce computational complexity when handling larger input feature map. In addition, we visualize the shallow feature and deep feature for ResNet-50 and weight encoder in Fig. 4 to visualize and interpret the rationality of the PE-free attention module in MMTB, as shown in Fig. 5. We can make the following observations.

1) Compared with ResNet-50, weight encoder extracts more effective information.
2) Weight encoder effectively produces precise global spatial information required by deep decoder in shallow features, while able to gradually generate the local positional information required by superficial decoder in

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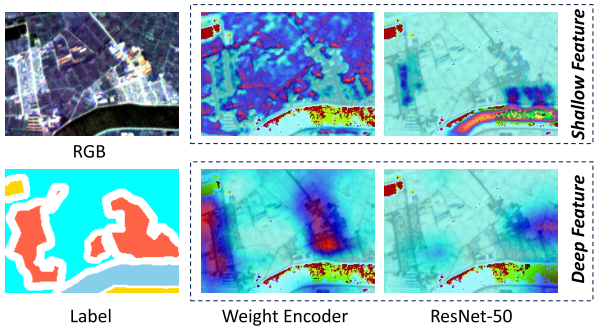


Fig. 4. Visualization of feature maps obtained by two encoders on rural settlement. (Top row) Shallow feature. (Bottom row) Deep feature.

deep features. The positional information guarantees the rationality of the PE-free module, and its effectiveness will be verified in Section III-D5.

D. Linear-Fuse Segmentation Head

The multiscale feature map generated during the decoding process exhibits inherent semantic differences. Effective feature fusion can enrich the semantic information and enhance the expressiveness of segmentation results. Considering that the Transformer translates 2-D image-based tasks into 1-D sequence-based tasks, an LFSH is used to realize the alignment of semantic information among feature maps in each stage, the details of which are described as follows.

Since the MMTB module outputs each feature map $X_j \in \mathbb{R}^{1\times 1\times HW}(j = 1, 2, 3, \ldots, 128)$, for simplicity let F be the overall feature map. Assume that $F_k \in \mathbb{R}^{128\times(HW/4^{|k-5|})}(k = 1, 2, 3, 4)$ is the feature map generated by the MMTB stage#k. As shown in Fig. 6, the operation process of the LFSH module can be decomposed into the following three steps. First, multiscale feature map F_k generated from multistage MMTB decoder go through multilinear layers to map different features into a unified segmentation space. Second, each level of features is assigned a learnable weight and sampled to a uniform spatial dimension. A linear fuse module is a 1×1 convolutional layer with a stride of 1 followed by GN, which is adopted to fuse features. The operation of concatenation before fusion can be weighted selectively according to the similarity between multiscale features and fused feature to realize the alignment of semantic information. Third, a 3×3 convolutional layer is employed to further refine the fused feature, and then a postprocessing 1×1 convolutional layer and an upsampling operation are applied to produce the final segmentation mask $M \in \mathbb{R}^{H\times W\times N_{\text{class}}}$, where N_{class} is the number of categories. During this step, we introduce residual connections to prevent network degradation. The above operations can be represented as follows:

$$F_k' = F_k \cdot W_k^T + b_k, \forall k \tag{13}$$

$$F' = \text{concat}\left(F_k' \odot (\alpha, \beta, \gamma, \eta)\right), \ \forall k \tag{14}$$

$$\hat{F}' = \delta\left(\text{GN}\left(\text{Conv1} \times 1\left(F'\right)\right)\right) \tag{15}$$

$$M = \text{Upsampling}\left(\text{Linear}\left(ED', N_{\text{Class}}\left(\text{Conv3} \times 3\left(\hat{F}'\right)\right)\right)\right) \tag{16}$$

where W_k, b_k denote the parameters and bias of multilinear layers, and $W_k \in \mathbb{R}^{\text{EmbedDim}\times 128}, b_k \in \mathbb{R}^{\text{EmbedDim}\times 1}$. $\odot$ represents the elementwise product. Finally, we use the cross-entropy loss function to calculate the training loss and update the model weights, which can be represented as follows:

$$L_{\text{CE}} = -\frac{\sum_{i=1}^{N_s} y^{(i)} \sum_{j=1}^{N_{\text{class}}} \log \hat{y}_j^{(i)}}{N_s} \tag{17}$$

where N_s represents the number of samples. $y^{(i)}$ is the one-hot encoding of the true semantic labels of sample i. $\hat{y}_j^{(i)}$ represent the confidence of sample i belonging to the category j.

At the end of this section, the pseudocode of the proposed UM2Former is shown in Algorithm 1.

Algorithm 1 U-Shaped Multimixed Transformer

Input: HSI trainset S_{train}: $\{X_{train}, Y_{train}\}$ and testset S_{test}: $\{X_{test}, Y_{test}\}$,, training epoch for UM2Former N_{epo}.

Output: Predicted label for HSI.

I. Training Stage:

for *i=1 to* N_{epo} **do**

  Feed each HSI data $I \in \mathbb{R}^{H\times W\times C}$ from X_{train} into UM2Former to get segmentation mask $M \in \mathbb{R}^{H\times W\times N_{class}}$, the prediction vector $\hat{Y}_{train}$ is formed by all segmation masks;

  Calculate L_{CE} using Eq.(17) by Y_{train} and $\hat{Y}_{train}$;

  Train UM2Former using L_{CE} with Adam optimizer.

end

II. Testing Stage:

Feed samples X_{test} from S_{test} into trained UM2Former to get predicted labels.

III. Experimental Results

A. Datasets

The WHU-OHS images are acquired by Orbita hyperspectral satellites (OHS) of Zhuhai-1 from more than 40 regions (by city) in China. The spatial resolution of WHU-OHS dataset is 10 m, including 32 spectral bands between 400 and 1000 nm. Each image is seamlessly cropped into the same image patches, and the size and spectral bands of these patches are $\{(512 \times 512), 32\}$. The training set, validation set, and test set of each OHS satellite image are all composed of image patches, with a ratio of 9:1:5. It is worth noting that the training set and test set are entirely isolated, and the validation set and training set have no intersection in space.

In this section, we evaluate the effectiveness of our model in HSI segmentation tasks using five typical cities located in Northeast, South, North, and Northwest China: Jilin (T1), Guangzhou (S3), Tianjin (S8), Karamay (T5), and Wuzhong (O21) as target areas. Each image contains 13 categories on average, and the training and testing datasets have the same feature distribution. The datasets selected for the experiments include all 24 categories divided by WHU-OHS data, and their corresponding ground object categories are presented in Table II. The number of image patches contained in the training, test, and validation sets, as well as the cities

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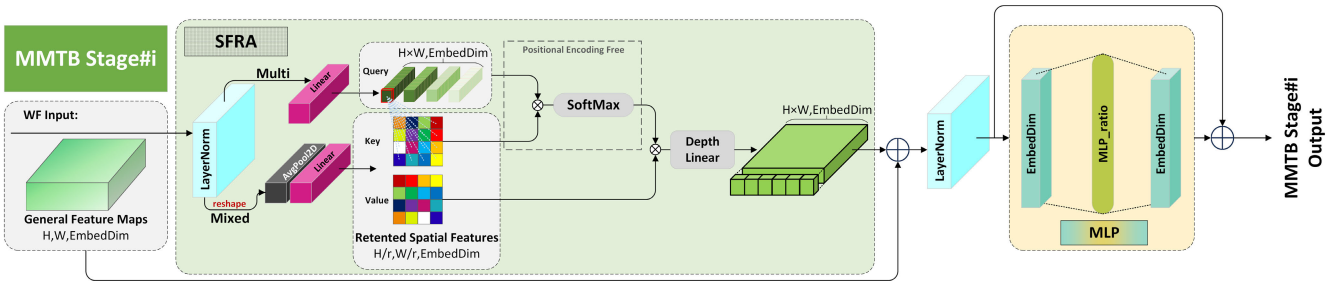


Fig. 5. Graphical illustration of the proposed MMTB.

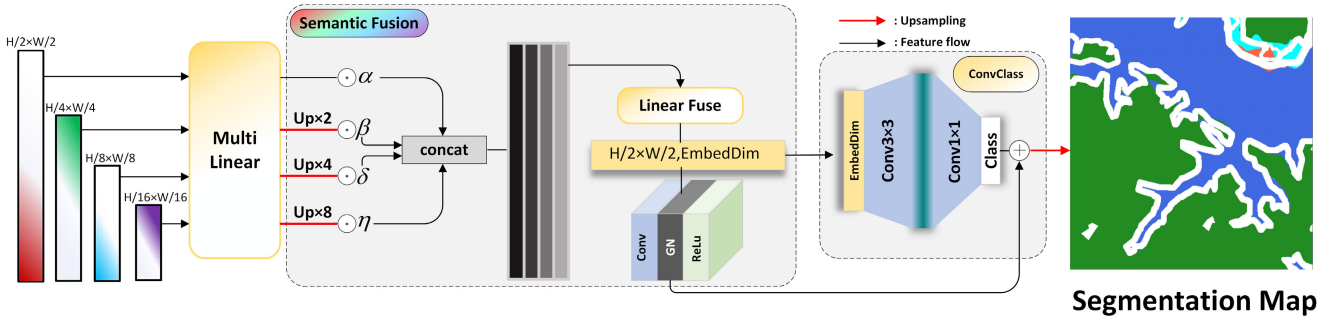


Fig. 6. Graphical illustration of the proposed LFSH.

TABLE II
CLASSES FOR FIVE EXPERIMENTAL AREAS

Major Class	Class	Experimental Areas				
		T1	S3	S8	T5	O21
Farmland	Paddy field	✓	✓	✓		✓
	Dry farm	✓		✓	✓	✓
Forest	Woodland	✓	✓			✓
	Shrubbery			✓		
	Sparse woodland		✓			
	Other forestland		✓	✓		✓
Grassland	High-covered Grassland		✓	✓	✓	✓
	Medium-covered Grassland			✓	✓	✓
	Low-covered Grassland			✓	✓	✓
Water	River/canal	✓	✓	✓		✓
	Lake	✓				✓
	Reservoir/pond	✓	✓	✓	✓	✓
	Beach land			✓		
	Shoal	✓	✓	✓		✓
Build-up	Urban build-up	✓	✓	✓	✓	✓
	Rural settlement	✓	✓	✓	✓	✓
	Other Construction land	✓	✓	✓	✓	✓
Unused land	Sand					✓
	Gobi				✓	✓
	Saline/Alkali soil				✓	
	Marshland	✓			✓	
	Bare land		✓	✓	✓	✓
	Bare rock	✓			✓	✓
Ocean	Ocean			✓		

they represent, are recorded in Table III. In addition, three-band false-color composite images and ground-truth maps corresponding to six images selected from test set of five target areas to visualize the segmentation effects of different models are shown in Fig. 7.

B. Experimental Settings

1) Hyperparameters: For fast convergence, we deploy the Adam optimizer to train our model in the experiments. The base learning rate is set to 1e-4, and the cosine strategy was employed to adjust the learning rate. For the training epoch, the training process would be stopped once the training loss is stabilized. Specifically, the training epoch is set to 250 for T1, 240 for S3, 230 for S8, 250 for T5, and O21. Considering the reliance of batch normalization on batch size, the computed mean and variance are unstable when the batch size is small. Given the limited training resources, we cannot apply a large batch; thus, we adopted the GN strategy in UM2Former. The batch and group sizes for all areas are set to 4 and 16. Other parameters, including learning rate, overlap patch size, reduction ratio in the channel-weight module, and number of heads in MMTB, will be optimized in the following experiments.

2) Evaluation Metrics: Two kinds of evaluation metrics are used in our experiment; the first ones are commonly used metrics for qualitative and quantitative evaluation of semantic segmentation performance, including the overall accuracy (OA), mean F1-score (mF1), and mean intersection over union (mIoU). OA, also known as pixel accuracy (PA), collects each pixel prediction

$$\mathrm{OA} = \frac{\sum_i \mathrm{TP}_i}{\sum_i (\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i)} \tag{18}$$

where TP_i is the number of true positives for category i, FP_i is the number of false positives for category i, and FN_i is the number of false negatives for category i. mF1 score is the harmonic mean of precision and recall, which is given by

$$F1_i = 2 \times \frac{\mathrm{precision}_i \times \mathrm{recall}_i}{\mathrm{precision}_i + \mathrm{recall}_i} \tag{19}$$

$$\mathrm{mF1} = \frac{1}{N} \sum_{i=1}^{N} F1_i \tag{20}$$

where N is the total number of categories. mIoU computes the intersection and union of two sets, which is defined by

$$\mathrm{IoU}_i = \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i} \tag{21}$$

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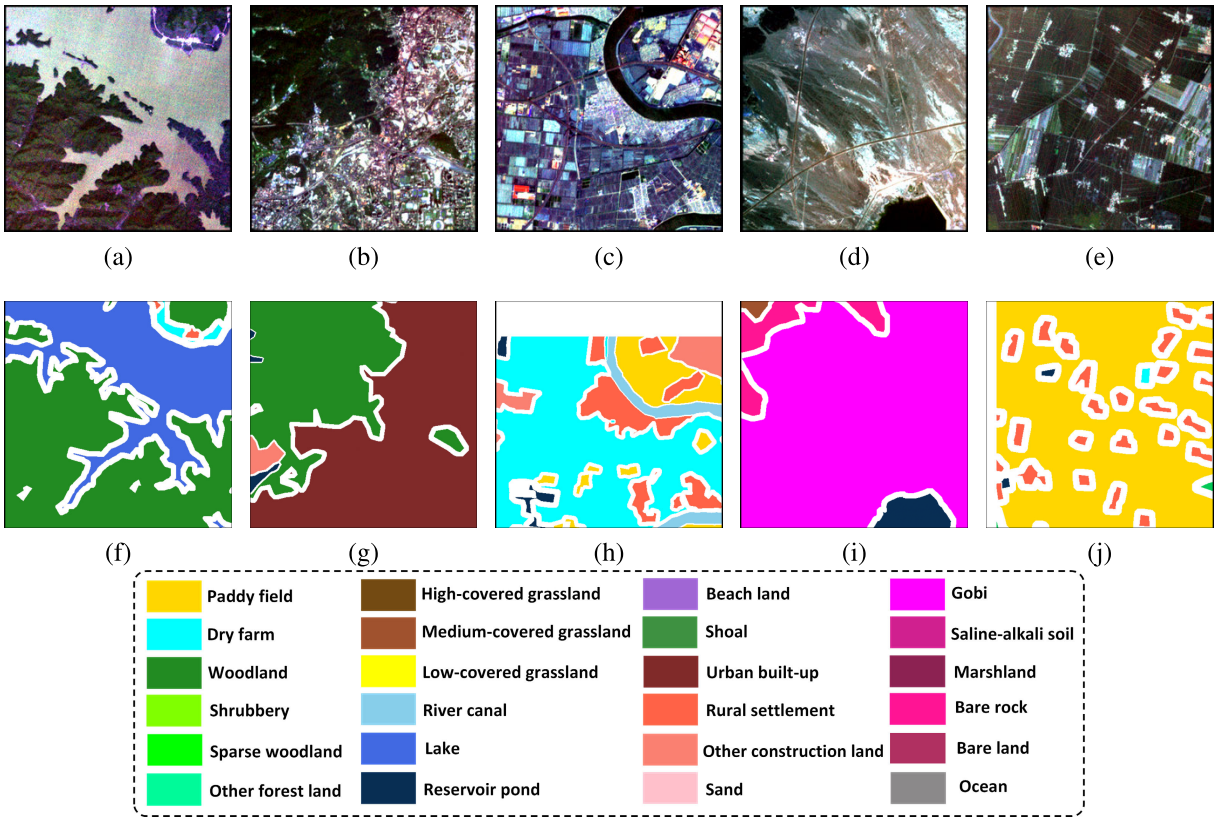


Fig. 7. Images selected for visualization from the test set (R: 2, G: 7, B: 14). (a) T1_0134. (b) S3_0037. (c) S8_0054. (d) T5_0023. (e) O21_0069. (f)–(j) Ground-truth map.

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{N} \text{IoU}_i. \tag{22}$$

The combination of OA, mF1, and mIoU ensures a comprehensive evaluation of the model performance. OA gives an overall accuracy of classification accuracy, mF1 balances precision and recall across all categories, and mIoU provides a detailed measure of segmentation performance, especially for smaller and more challenging categories.

And the second one is to evaluate the scale of the network, including the frames per second (FPS) to evaluate the speed, the floating-point operations (FLOPs) to measure the complexity of the method, and parameters size (MB) to evaluate the memory requirement. All the experiments were implemented on environment with an Intel Core 13600KF CPU, a NVIDIA GeForce RTX 3090 GPU with 24 GB memory and 32 GB RAM, based on PyTorch 1.12.0 and cuDNN 11.3 framework. The results of the experiments are the best results presented.

3) Models for Comparison: We compared the proposed UM2Former with the typical DL methods for HSI classification: 3DCNN [54] and attention-based adaptive spectral–spatial kernel ResN et (A^2S^2K-ResNet) [55]), CNN-based networks for semantic segmentation, including patch-free global learning framework (FreeNet) [39] and unified perceptual parsing network (UperNet) [58], and Tansformer-based networks for semantic segmentation (DCSwin [30], multiattention network (MANet) [59], TransUNet [60], U-Net like Transformer (U-NetFormer) [48], hierarchical ViT (HiViT) [61], UCTransNet [62], TransXNet [63], and SegFormer [47]). The methods in this article are all implemented by PyTorch. To assess the segmentation performance of each method, the evaluation metrics used are OA, mF1, and mIoU. These metrics provide a comprehensive measure of the accuracy and agreement between tested and ground-truth labels.

1) In the typical DL methods for HSI classification networks, the iterations are executed to 150 for all datasets, the batch size is set to 4, and learning rate is fixed at 0.001.
2) In the CNN-based networks for semantic segmentation networks, the feature extractor component of UperNet utilizes the ResNet-50 architecture. The segmentation head of UperNet includes three feature fusion layers and the number of feature maps extracted by the encoder of FreeNet is 64. In addition, the iterations are executed to 200 for all datasets, the batch size is set to 8, and learning rate of UperNet is fixed at 0.0001, while the learning rate of FreeNet is set to 0.00005.
3) In the Tansformer-based networks for semantic segmentation networks, for fast convergence, we employ the Adam optimizer in the experiment for training these models and adjusted the learning rate during training through a cosine annealing strategy. For U-NetFormer, DCSwin, and MANet, the learning rate is fixed to 0.0005, whereas for SegFormer and HiViT it is set to 0.0001. The learning rate of TransXNet, TransUNet, and UCTransNet is 0.001. In addition, the batch size of these compared methods is set to 8 and the iterations are executed to 250. The number of feature maps extracted

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TABLE III
NUMBER OF TRAINING, TEST, AND VALIDATION PATCHES FOR EACH CITY

Image	City	Number of Image Patches		
		Training	Test	Validation
T1	Jilin	120	58	13
T2	Shanghai	126	41	13
T3	Zhongshan	120	37	13
T4	Lanzhou	113	55	12
T5	Kelamayi	108	62	11
T6	Nanchang	114	51	12
T7	Changsha	122	69	13
T8	Tangshan	105	51	11
S1	Changchun	110	74	12
S2	Wuxi	107	51	11
S3	Guangzhou	113	63	12
S4	Xining	106	36	11
S5	Hetian	108	68	12
S6	Anyi	112	56	12
S7	Changde	122	66	13
S8	Tianjin	114	63	12
O1	Baoding	106	36	11
O2	Chongqing	117	68	12
O3	Fujin	110	55	12
O4	Huainan	109	63	12
O5	Huhehaote	136	62	15
O6	Jinzhong	118	71	13
O7	Lvliang	124	62	13
O8	Manasi1	127	62	14
O9	Manasi2	115	72	12
O10	Nanmulin	111	52	12
O11	Neimenggu	113	45	12
O12	Qingdao	120	64	13
O13	Qinghuangdao	112	66	12
O14	Shawan	124	64	13
O15	Shenyang	108	70	12
O16	Shuozhou	115	47	12
O17	Songpan	113	58	12
O18	Taian	117	61	12
O19	Tongjiang1	115	46	12
O20	Tongjiang2	125	73	13
O21	Wuzhong	112	80	12
O22	Xundian	112	55	12
O23	Xuzhou	108	61	11
O24	Yidu	112	42	12
O25	Zangzu	104	62	11
O26	Zhongshan	118	61	13
Total	42	4822	2460	513

by the encoder of U-NetFormer and SegFormer is 64 and 256, respectively.

C. Parameter-Sensitive Analysis

1) Learning Rate: To determine the appropriate learning rate for model training, we sequentially set the learning rate within the range of [1e-2, 5e-3, 1e-3, 5e-4, 1e-4, 5e-5, 1e-5]. Subsequently, we conduct experiments on the aforementioned five regions and record the mIoU score achieved by the model under each learning rate configuration. As shown in Fig. 8(a), when the learning rate ranges from 1e-2 to 1e-4, the mIoU obtained by UM2Former exhibits a consistent upward trend on S3, S8, T5, O21, and 1e-2 to 5e-4 on T1. However, mIoU shows an obvious decline when the learning rate falls below 1e-4 on S3, S8, T5, O21, and 5e-4 on T1. Based on the experimental results, we set the learning to 1e-4 for S3, S8, T5, O21, and 5e-4 for T1.

2) Overlap Patch Size: In the experiment, the impact of overlap patch size on segmentation accuracy is evaluated. We set overlap patch size to [1, 2, 3, 4, 7, 8, 16], respectively. As shown in Fig. 8(b), when the patch size is set to 4×4, the highest mIoU can be achieved compared to other configurations for T1, S3, and O21. For S8 and T5, the highest mIoU can be achieved when the patch size is set to 8×8. Based on the experimental results, we set the patch size to 4×4 for T1, S3, and O21, and 8×8 for S8 and T5.

3) Reduction Ratio of Channel-Weight Module: To ensure model simplicity and universality, we set reduction ratio of the dimensionality reduction layer in channel-weight to [2, 4, 8, 16, 32, 64]. The weighted feature maps generated by this module are then fed into MMTB, and the resulting impact on segmentation accuracy is analyzed across different feature maps. As shown in Fig. 8(c), UM2Former obtains higher segmentation accuracy than other parameter configuration when reduction ratio is set to 16 for all five regions.

4) Number of Head in MMTB Block: Number of heads determine the degree of information interactions between query vector and all key-value vectors, and also determine the parameter size of the decoder. We set this parameter to [1, 2, 4, 8, 16, 32], respectively, and compare segmentation performance of the model under different settings. As shown in Fig. 8(d), UM2Former achieves higher segmentation accuracy than other configuration on five regions when this parameter is set to 8.

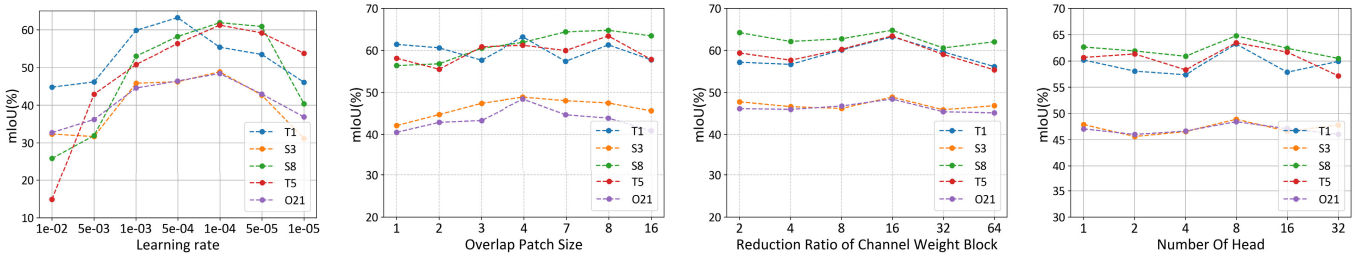
D. Ablation Study

1) Effectiveness of Weight Encoder: The weight encoder is the core module of UM2Former for feature extraction, corresponding to overlap-down and channel-weight modules. We first extract the overlap-down and channel-weight modules in this part to validate their effectiveness. And then, to demonstrate the advantages of the proposed efficient weight encoder in HSI segmentation task, we replace it with ResNet-50 [64], SwinTransformer [26], and HiViT [61], which are commonly used as encoders in many segmentation methods.

As reported in Table IV, using weight encoder can effectively improve segmentation accuracy compared with not using them, which indicates that our design can effectively extract the discriminative spectral–spatial feature maps required for HSI segmentation tasks. In addition, we observed that using the overlap-down module yields a higher mIoU in T1, S3, S8, and O21 than using the channel-weight module only. This is because the overlap-down module enables the extraction of uninterrupted local spatial information while simultaneously expanding the receptive field. Using only channel-weight module, the pooling layer will lead to a loss of spatial details and, consequently, poorer results. Furthermore, compared to ResNet-50, employing weight encoder as encoder yields a higher mIoU. Especially, mIoU improved by 13.04% for T5 and 11.58% for O21, respectively.

Current Transformer-based HSI segmentation networks typically use Transformer as the encoder. Although this choice is reasonable for extracting global semantic information, it will greatly reduce the execution speed of the network. Meanwhile, segmentation network based on CNN is unable to achieve efficient segmentation. To demonstrate it, we replace our

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(a) (b) (c) (d)

Fig. 8. Evolution of mIoU as a function of (a) learning rate, (b) overlap patch size, (c) reduction ratio of channel-weight module, and (d) number of heads.

TABLE IV
ABLATION STUDY OF WEIGHT ENCODER IN FIVE REGIONS

Weight-encoder		mIoU(%)				
Overlap-Down	Channel-Weight	T1	S3	S8	T5	O21
✓	✗	61.21	47.12	61.91	59.22	45.17
✗	✓	57.92	45.93	58.99	62.16	44.56
✓	✓	**63.22**	**48.87**	**65.71**	**63.44**	**48.41**

weight encoder with lightweight Transformers, i.e., Swin-Tiny and HiViT-Base, and lightweight CNN, ResNet-50, for ablation studies. According to Table V, the results reveal that using Swin-Tiny as encoder provides improvement of mIoU by 0.7% for T1, 0.95% for S3, 1.79% for T5, and 1.1% for O21, but the complexity and the parameter size of the model are greatly increased. These results further demonstrate the effectiveness of the proposed Transformer-based decoder and its potential in a fully Transformer structure. In addition, compared to ResNet-50, employing weight encoder as encoder yields a higher mIoU. Especially, mIoU improved by 13.04% for T5 and 11.58% for O21, respectively. Thus, for large-scale HSI semantic segmentation, the application of a lightweight encoder like weight encoder is efficient and effective.

2) Effectiveness of WF Operation: To evaluate the effectiveness of the WF operation, we conduct separation experiments by replacing the WF operation with a direct addition of features extracted from the encoder and decoder. As reported in Table VI, the WF operation significantly enhances the accuracy of the weight encoder across all five regions. For instance, the mIoU increased by 6.08% for T1, 2.17% for S3, 1.30% for S8, 4.85% for T5, and 2.14% for O21. Furthermore, we replace the weight encoder with ResNet-50 and performed the same experiment to confirm the applicability of the WF operation, yielding similar results. Specifically, mIoU improved by 0.69% for T1, 0.61% for S3, 2.02% for S8, 1.13% for T5, and 0.63% for O21. In the proposed UM2Former, the WF operation integrates two features based on their contribution to segmentation accuracy through learnable parameters, thereby facilitating the learning of more generalized fusion features.

3) Effectiveness of LFSH: In the overall structure of the coder and decoder (Codec), another essential component is segmentation head, which corresponds to linear-fusion segmentation head (LFSH) in the proposed UM2Former. In this part, we first conduct ablation experiments by replacing the semantic fusion module in LFSH with direct fusion to verify the effectiveness of this module. As reported in Table VII, using semantic fusion module can effectively improve segmentation accuracy compared with directly fusing features extracted by hierarchical decoder. This is because semantic fusion module can enhance the hierarchical features through learnable parameters and reduce their similarity to the fused feature, facilitating the interaction of semantic information between different layers and precluding the model from relying on a single feature. In addition, we compare the accuracy achieved by different segmentation head, including the widely employed CNN segmentation head in RS semantic segmentation, along with two specially designed segmentation heads: feature refinement segmentation head (FRSH) implemented in U-NetFormer and LFSH applied in UM2Former. The results outline the segmentation accuracy achieved by different segmentation heads when integrated into UM2Former backbone. Notably, LFSH achieves the highest accuracy. We believe that employing LFSH in constructing the segmentation head could align semantic information among multiscale feature maps. The mIoU obtained by CNN method and LFSH is reported in Table VII, the LFSH enhances the accuracy of the UM2Former across all five regions. Specifically, mIoU increases by 2.86% for T1, 1.24% for S3, 1.65% for S8, 3.87% for T5, and 1.24% for O21, respectively.

4) Effectiveness of MMTB: To demonstrate the advantages of the proposed SFRA, we replaced it with other advanced attention mechanisms to reconstruct the variants of UM2Former for ablation study. Benefiting from modeling the global dependency of each pixel in the weighted feature maps with the spatial retention feature, the deployment of our SFRA mechanism achieves the highest mIoU in all variant structure experiments on the T1, S3, S8, and T5, as listed in Table VIII. Experimental results indicate that the proposed MMTB is more suitable for performing as a decoder for processing feature maps with precise spatial information than other effective Transformer modules. In addition, we compare the computational complexity and the size of parameters of the proposed SFRA with other attention modules. Specifically, we compare the FLOPs (G) and parameter (M) of different decoders refactored by different attention mechanisms. SFRA achieves the highest accuracy among all variants, with a smaller model parameter size than other effective attention mechanisms in Transformer-based decoders.

5) Effectiveness of PE-Free Module: In this experiment, we analyze the effect of removing the PE in Transformer decoder. To this end, we train MMTB by using and excluding PE modules, respectively. Table IX shows the results for

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TABLE V
ABLATION STUDY OF DIFFERENT ENCODERS IN FIVE REGIONS

Encoder	Backbone	Decoder	Scale		mIoU(%)				
			FLOPs(G)	Parameter(M)	T1	S3	S8	T5	021
ResNet [64]	ResNet-50	MMTB	23.70	14.7	59.01	45.37	59.21	54.17	36.83
SwinTransformer [26]	Swin-Tiny		69.20	66.57	**63.92**	**49.82**	64.43	**65.23**	**49.42**
HiViT [61]	HiV-Base		111.75	68.13	53.46	39.81	55.47	55.68	39.86
Weight-encoder	-		39.52	6.77	63.22	48.87	**65.71**	63.44	48.41

TABLE VI
ABLATION STUDY OF CONNECTION LAYER IN FIVE REGIONS

WF	Encoder Block		mIoU(%)				
	Weight-encoder	ResNet-50	T1	S3	S8	T5	O21
✗	✓	✗	57.14	46.70	64.41	58.59	46.27
✓	✗	✓	59.01	45.37	59.21	54.17	36.83
✗	✗	✓	58.32	44.76	57.19	49.27	36.20
✓	✓	✗	**63.22**	**48.87**	**65.71**	**63.44**	**48.41**

TABLE VII
ABLATION STUDY OF SEGMENTATION HEAD IN FIVE REGIONS

Segmentation Head			mIoU(%)				
LFSH	CNN	FRSH	T1	S3	S8	T5	O21
✗	✓	✗	60.36	47.63	64.06	59.57	47.17
✗	✗	✓	53.51	40.56	57.86	58.67	42.41
Directly Fusion	✗	✗	56.47	45.66	63.41	60.50	46.91
Semantic Fusion	✗	✗	**63.22**	**48.87**	**65.71**	**63.44**	**48.41**

PE experiments. As shown, the proposed MMTB without PE module clearly outperforms using it. Taking mIoU as an example, it increased by 4.57% for T1, 0.69% for S3, 7.77% for S8, 4.48% for T5, and 3.82% for O21, respectively.

E. Segmentation Results

To verify the superiority of UM2Former in the context of large-scale HSI semantic segmentation, the accuracy results of different models are reported in Tables X–XIV for T1 (Jilin), S3 (Guangzhou), S8 (Tianjin), T5 (Kelamayi), and O21 (Wuzhong), respectively. The best segmentation accuracies are highlighted in boldface letters. These results reveal that the proposed UM2Former surpasses other methods on three metrics: OA, mF1, and mIoU.

1) HSI Classification Methods: UM2Former exhibits superior performance compared with the traditional HSI classification methods. For T1 (Jilin) dataset, our method outperforms the 3DCNN and A^2S^2K-ResNet with an average increase of 1.78% in OA and 25.50% in mIoU. Similarly, for the S3 (Guangzhou) dataset, UM2Former achieves OA and mIoU that are, on average, 4.77% and 22.97% higher than these HSI classification methods. In addition, our method achieves an average improvement of 4.87% in OA and 32.49% in mIoU on the S8 (Tianjin) dataset compared to 3DCNN and A^2S^2K-ResNet, and on the T5 (Kelamayi) dataset, with an average improvement of 5.55% in OA and 18.33% in mIoU. Finally, for the O21 (Wuzhong) dataset, UM2Former is beyond these HSI classification methods by 7.035% and 22.49% in terms of OA and mIoU, on average. This improvement can be attributed to the ability of our method to extract discriminative spectral–spatial feature maps through the establishment of hierarchical structured network, which empowers the algorithm to capture intricate patterns and nuances, thereby enhancing segmentation performance. Although HSI classification methods like 3DCNN and A^2S^2K-ResNet can also extract effective spectral–spatial features, the inputs of these HSI classification methods are overlapping patches generated in windows of small size (e.g., 9×9 for A^2S^2K-ResNet), the limited receptive fields of these methods could potentially restrict their capacity to capture generalized features from large-scale HSI datasets, which may affect classification performance. With regard to HSI classification algorithms, it can be seen in Tables X–XV that the proposed UM2Former achieves the best performance without introducing too many extra parameters and FLOPs.

2) Semantic Segmentation Convolutional Networks: When it comes to semantic segmentation convolutional networks, methods that rely solely on convolutional operations (e.g., UperNet) in encoder and decoder structure generally produce lower results than embedded attention mechanism (e.g., FreeNet and UM2Former). CNN-based methods can extract rich local texture information of HSI to distinguish subtle differences between different land cover categories. In contrast, spectral attention utilizes the global spatial context to model the interdependencies of the feature maps. We consider that since the spatial resolution of WHU-OHS dataset is 10 m and it has higher spectral resolution than natural and fine-resolution semantic segmentation datasets, it is easier to achieve higher segmentation accuracy for UM2Former and FreeNet based on spectral attention mechanism, which can utilize redundant spectral information more efficiently than CNN architecture models. In Table X, the results on the T1 dataset show that UM2Former outperforms CNN-based methods by 3.80% in OA and 11.64% in mIoU, on average. Similarity, in Table XI, the results on the S3 dataset indicate that our method achieves an average improvement of 4.83% in OA and 8.34% in mIoU. In addition, in Tables XII and XIII, our method achieves an average improvement of 1.97% in OA and 16.19% in mIoU on the S8 dataset compared to FreeNet and UperNet, and on the T5 dataset, with an average improvement of 4.27% in OA and 11.62% in mIoU. Finally, in Table XIV, the results on the O21 dataset demonstrate that UM2Former surpasses CNN architecture methods by 6.88% in OA and 8.89% in mIoU, on average. These findings highlight the advantage of UM2Former, which considers both redundant spectral information and global spatial information, leading to the improved performance on the evaluated large-scale HSI dataset.

3) Semantic Segmentation Transformer Methods: When considering both the construction of the Transformer-based decoder and the effect of PE interpolation, UM2Former

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TABLE VIII
ABLATION STUDY OF DECODER IN FIVE REGIONS

Decoder	Parameter(M)	FLOPs(G)	Encoder Block		Segmentation Head		mIoU(%)				
			Weight-encoder	ResNet-50	LFSH	CNN	T1	S3	S8	T5	O21
Shifted window attention (Liu *et al.*, 2021 [26])	10.63	46.77	✓	✗	✓	✗	52.32	43.11	45.28	50.53	38.71
			✓	✗	✗	✓	51.40	39.88	43.17	56.29	36.47
			✗	✓	✓	✗	52.52	43.30	52.91	52.32	38.02
			✗	✓	✗	✓	51.88	40.10	45.96	52.67	43.32
Global-local attention (Wang *et al.*, 2022 [48])	2.34	15.32	✓	✗	✓	✗	53.58	41.09	41.71	51.09	40.53
			✓	✗	✗	✓	52.28	38.66	39.65	56.16	34.10
			✗	✓	✓	✗	56.75	41.83	52.63	50.63	39.00
			✗	✓	✗	✓	54.16	40.34	42.45	54.71	43.49
Linear attention (Li *et al.*, 2021 [59])	1.54	**8.96**	✓	✗	✓	✗	58.84	44.27	55.21	59.64	41.52
			✓	✗	✗	✓	50.75	36.79	39.51	51.39	34.20
			✗	✓	✓	✗	55.76	43.54	48.52	52.73	37.87
			✗	✓	✗	✓	48.96	35.96	38.46	44.50	27.24
Spatial-feature-retention attention (our)	**1.14**	14.46	✓	✗	✓	✗	**63.22**	**48.87**	**65.71**	**63.44**	**48.41**
			✓	✗	✗	✓	60.36	47.63	64.06	59.57	47.17
			✗	✓	✓	✗	59.01	45.37	59.21	54.17	36.83
			✗	✓	✗	✓	58.46	46.08	59.63	58.22	37.64

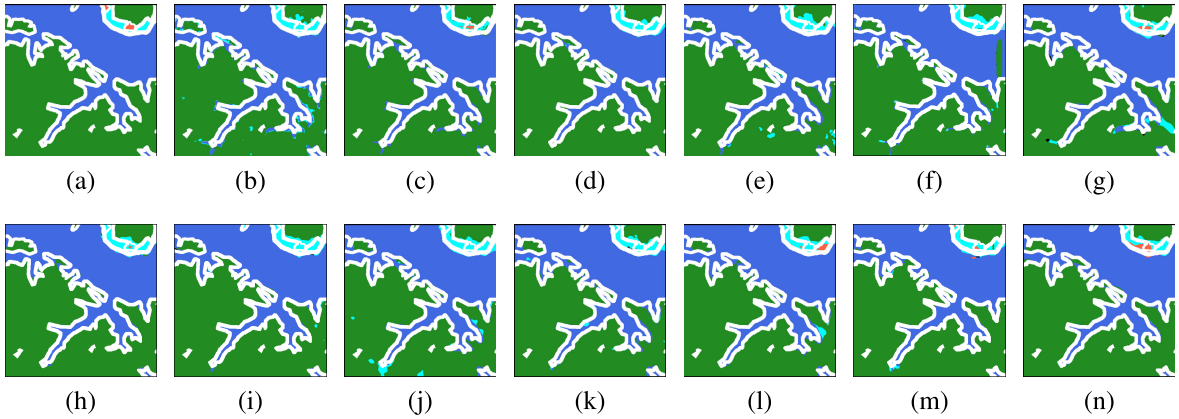


Fig. 9. Visualization results from the T1 test set. (a) Ground truth. (b) 3DCNN (36.02%). (c) A^{2}S^{2}K-ResNet (39.41%). (d) FreeNet (50.05%). (e) UperNet (53.11%). (f) TransUNet (37.54%). (g) U-NetFormer (43.83%). (h) HiViT (44.01%). (i) TransXNet (51.31%). (j) DCSwin (47.88%). (k) MANet (48.14%). (l) SegFormer (50.40%). (m) UCTransNet (50.06%). (n) UM2Former (63.22%).

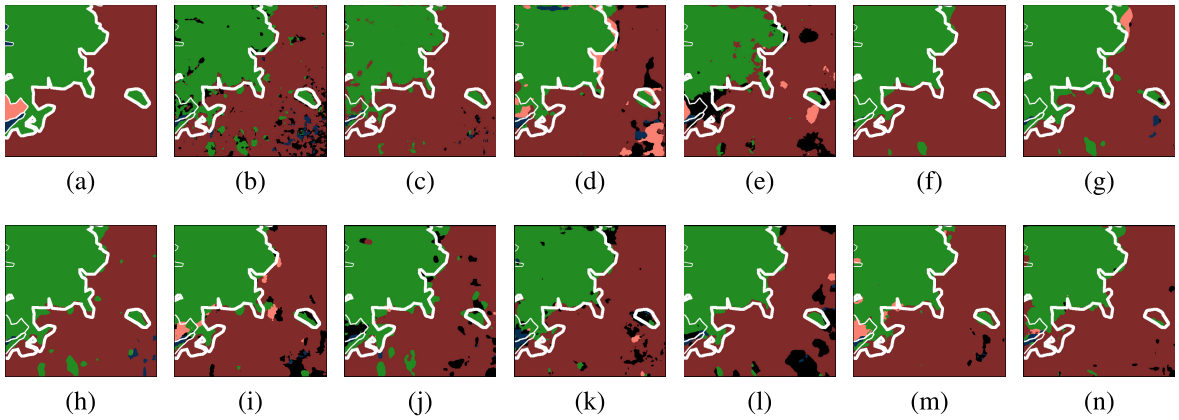


Fig. 10. Visualization results from the S3 test set. (a) Ground truth. (b) 3DCNN (22.14%). (c) A^{2}S^{2}K-ResNet (29.65%). (d) FreeNet (42.15%). (e) UperNet (38.90%). (f) TransUNet (25.75%). (g) U-NetFormer (29.23%). (h) HiViT (31.95%). (i) TransXNet (36.92%). (j) DCSwin (30.32%). (k) MANet (35.25%). (l) SegFormer (33.08%). (m) UCTransNet (35.80%). (n) UM2Former (48.87%).

outperforms U-NetFormer, which only focuses on the construction of the Transformer-based decoder. In addition, HiViT, DCSwin, MANet, and SegFormer focus on employing Transformer blocks as the encoder, this design can effectively extract global information while losing spatial details to some extent. Furthermore, the self-attention mechanism of TransUNet and UCTransNet cannot effectively integrate spectral–spatial information, which may lead to inaccurate segmentation of certain details. In contrast, UM2Former develops MMTB as a decoder to process feature maps with abundant spectral–spatial information and models the global dependence of each pixel in the feature maps from encoder using spatially preserved features through the SFRA. The advantages can be observed from the results of the T1 dataset in Table X, where our method achieves 1.24% and 15.62% higher OA and mIoU on average compared to

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TABLE IX
ABLATION STUDY OF PE MODULE IN FIVE REGIONS

Region	PE or Not	mIoU(%)	mF1(%)
T1	Y	58.65	69.66
	N	**63.22**	**74.52**
S3	Y	48.18	61.44
	N	**48.87**	**61.98**
S8	Y	57.94	69.83
	N	**65.71**	**76.95**
T5	Y	58.96	68.95
	N	**63.44**	**73.80**
O21	Y	44.99	74.01
	N	**48.41**	**75.27**

the semantic segmentation Transformer methods. Similarly, in the S3 dataset (Table XI), UM2Former produces an average improvement of 4.31% in OA and 16.08% in mIoU. In addition, in Tables XII and XIII, our method achieves an average improvement of 1.78% in OA and 25.92% in mIoU on the S8 dataset, and on the T5 dataset, with an average improvement of 3.47% in OA and 16.31% in mIoU. Finally, in Table XIV, the results on the O21 dataset demonstrate that UM2Former surpasses other Transformer-based methods by 5.97% in OA and 14.65% in mIoU, on average. These findings further demonstrate the effectiveness of the proposed Transformer-based decoder.

F. Visualization Results

To reflect the segmentation performance of different models visually, the ground truth and segmentation maps based on different models for five selected test images are displayed respectively in Figs. 9–14.

1) T1 (Jilin): The corresponding segmentation maps produced by different models are shown in Fig. 9, and the segmentation maps generated by UM2Former show better homogeneity within areas representing the same land cover categories like *Shoal* and *Lake*. In addition, UM2Former achieves the highest segmentation accuracy in 4 of 12 land cover categories in T1. Specifically, the *Bare rock* category was distinguished solely by UperNet, MANet, and our method achieving the highest IoU of 84.83%.

2) S3 (Guangzhou): UM2Former achieves highest category segmentation score in 7 of 12 land cover categories in S3. Specifically, only three methods have successfully distinguished the *Shoal* category, and our method achieves the highest IoU score of 77.69%. The segmentation maps are shown in Fig. 10. The segmentation map produced by UM2Former demonstrates superior homogeneity in identifying *Urban build-up* and *Woodland* categories compared to other models.

3) S8 (Tianjin): UM2Former achieves the highest IoU score in 6 of 16 land cover categories, especially for *Shrubbery* category in which UM2Former achieved highest segmentation accuracy while other models failed to recognize this category correctly. The segmentation maps, as shown in Fig. 11, benefiting from the pixelwise construction of global contextual information by MMTB, UM2Former can model the global positional relationship while distinguishing the differences between different land cover categories, thus achieving more accurate and smoother segmentation results like *Rural settlement* and *River canal*.

4) T5 (Kelamayi): In terms of class-specific accuracy, UM2Former still has higher performance and makes strong enough predictions in some challenging categories such as *Other Construction land* and *Bare land*, which evidences the robustness of our UM2Former. As show in Fig. 12, the segmentation map of UM2Former is the smoothest in *Reservoir pond* and *Gobi*, while other methods have some misrecognized pixels.

5) O21 (Wuzhong): In terms of class-specific accuracy, UM2Former has the highest IoU score among the six classes and makes strong enough predictions in some challenging categories such as *Woodland* and *Bare land*, which evidences the robustness of our method. The segmentation maps, which are shown in Fig. 13. It can be found that in the comparison methods, all have different degrees of misclassification phenomenon, showing unsmooth or even distorted areas. For example, part of the *Rural settlement* is wrongly identified as *Paddy field*. The reason for this phenomenon is that CNN ignores irrelevant information of the image and Transformer methods based on partitioning patches lack effective attention to pixel-level semantic information. However, UM2Former presents more smooth and correct segmentation results benefiting from the pixelwise construction of global contextual information by MMTB.

6) Disparity Analysis of Evaluation Metrics Across Methods: In this part, we analyze the differences observed in different comparison methods under different precision metrics. On the one hand, in semantic segmentation convolutional networks, such as FreeNet, tend to exhibit a balanced performance between OA and mIoU. For example, as shown in Table XI, FreeNet achieves the second highest mIoU of 42.15% and has a slightly lower OA of 77.82% than UM2Former. This balance can be attributed to the design of convolutional architectures, which can distinguish subtle differences between different land cover categories and extract richer local texture information of HSI than the Transformer architecture model. As a result, high-performance convolutional models, like FreeNet and UperNet, are generally superior to Transformer-based models when dealing with smaller or more complex land cover classifications in different regions. On the other hand, Transformer-based methods highlight a stronger focus on global dependencies and gradually lose detailed features in the process of downsampling. This issue is especially pronounced in HSI, which is often constrained by limited spatial resolution. As a result, these methods generally achieve higher OA, but mIoU varies significantly. For instance, HiViT achieves a high OA but a lower mIoU, suggesting that it performs well in dominant categories but struggles with challenging ones. In contrast, benefiting from the local fine-grained features extracted by the weight encoder and the fine global modeling capability of SFRA, UM2Former shows better generalization ability in all categories. This improvement is particularly noticeable in smaller and more challenging land cover classes, such as *Bare rock*, *Shoal*, *Shrubbery*, and *Bare land*, where UM2Former significantly outperforms other methods.

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TABLE X
SEGMENTATION ACCURACY OF DIFFERENT METHODS IN T1

Methods	Backbone	IoU(%) Score per Category												OA(%)	mF1(%)	mIoU(%)
		Paddy field	Dray farm	Wood land	Rival canal	Lake	Reservoir	Shoal	Urban build-up	Rural settle-ment	Other construc-tion land	Marsh land	Bare Rock			
Hyperspectral image classification methods																
3DCNN [54]	/	35.08	60.95	92.97	74.58	94.73	0.00	0.00	69.58	4.41	0.00	0.00	0.00	89.46	41.44	36.02
A^{2}S^{2}K-ResNet [55]	/	41.62	66.40	95.46	79.98	95.39	0.00	0.00	73.56	20.58	0.00	0.00	0.00	91.32	45.14	39.41
Semantic segmentation Convolutional networks																
FreeNet [39]	/	48.73	66.23	95.02	79.00	**97.56**	22.03	31.50	86.33	**39.97**	23.86	10.39	0.00	91.93	60.07	50.05
UperNet [58]	ResNet-50	35.36	58.33	91.63	69.95	93.35	25.31	64.64	71.13	18.85	37.54	4.85	66.32	88.61	64.81	53.11
Semantic segmentation Transformer methods																
TransUNet [60]	ViT-B/16	41.40	65.43	94.52	61.66	96.00	0.00	0.00	77.04	18.37	0.00	0.00	0.00	90.26	43.74	37.54
U-NetFormer [48]	ResNet-18	41.39	63.94	94.59	74.36	94.18	11.25	29.49	68.41	23.33	25.07	0.00	0.00	90.38	53.42	43.83
HiViT [61]	HiViT-B	**50.41**	**69.59**	**96.52**	81.18	97.28	0.00	0.00	86.93	39.71	0.00	0.00	0.00	91.86	49.83	44.01
TransXNet [63]	TransX-T	42.88	68.23	95.37	82.00	96.64	28.17	44.98	**88.66**	33.97	**47.29**	0.00	0.00	92.04	60.88	51.31
DCSwin [30]	Swin-Tiny	31.26	57.74	92.89	76.53	94.58	17.08	64.03	75.54	24.98	39.96	0.00	0.00	89.21	57.63	47.88
MANet [59]	ResNet-50	38.57	65.80	95.45	76.14	96.00	7.46	28.43	87.04	35.50	16.59	0.60	30.08	91.57	58.04	48.14
SegFormer [47]	MIT-B1	42.02	61.45	93.67	76.65	95.57	16.55	**65.10**	77.76	33.01	43.07	0.00	0.00	90.50	60.09	50.40
UCTransNet [62]	/	41.17	68.94	96.27	77.58	96.75	19.63	48.17	83.96	36.83	31.36	0.00	0.00	91.98	59.54	50.06
UM2Former (Ours)	MMT	48.06	67.27	95.18	**87.00**	97.19	**28.80**	64.30	85.43	37.48	31.04	**36.57**	**84.83**	**92.17**	**74.52**	**63.22**

TABLE XI
SEGMENTATION ACCURACY OF DIFFERENT METHODS IN S3

Methods	Backbone	IoU(%) Score per Category												OA(%)	mF1(%)	mIoU(%)
		Paddy field	Wood land	Sparse wood land	Other forest land	High covered grassland	Rival canal	Reservoir	Shoal	Urban build-up	Rural settle-ment	Other Con-struction land	Bare land			
Hyperspectral image classification methods																
3DCNN [54]	/	45.20	62.03	0.00	0.00	0.00	70.14	8.43	0.00	67.32	0.00	12.50	0.00	70.31	28.29	22.14
A^{2}S^{2}K-ResNet [55]	/	61.78	72.39	0.00	14.31	5.32	78.96	21.39	0.00	77.45	5.63	22.68	0.00	78.54	37.07	29.65
Semantic segmentation convolutional networks																
FreeNet [39]	/	60.63	72.77	35.06	24.11	**56.43**	80.00	**21.00**	4.40	76.85	10.11	29.38	33.66	77.82	54.80	42.15
UperNet [58]	ResNet-50	43.35	64.00	29.25	23.89	32.86	71.73	19.25	37.05	68.59	10.00	22.14	44.74	70.92	53.27	38.90
Semantic segmentation Transformer methods																
TransUNet [60]	ViT-B/16	59.22	63.63	0.00	0.00	0.00	77.27	4.56	0.00	78.45	0.00	25.89	0.00	76.69	31.42	25.75
U-NetFormer [48]	ResNet-18	53.53	63.78	0.00	25.00	0.00	79.70	20.66	0.00	75.94	11.03	23.93	0.00	77.17	37.65	29.23
HiViT [61]	HiViT-B	**65.39**	73.49	0.00	26.85	0.00	80.28	20.94	0.00	79.55	0.00	**30.09**	0.00	78.29	39.77	31.95
TransXNet [63]	TransX-T	50.46	72.80	29.36	**27.74**	36.41	81.38	16.39	0.00	71.34	13.46	24.95	17.69	75.41	49.04	36.92
DCSwin [30]	Swin-Tiny	42.83	61.55	29.31	16.11	29.58	74.44	13.97	0.00	69.62	7.65	18.73	0.00	70.37	41.05	30.32
MANet [59]	ResNet-50	54.42	68.53	21.50	15.27	27.41	75.22	22.27	0.00	76.38	11.04	26.31	23.68	76.69	47.18	35.25
SegFormer [47]	MIT-B1	44.06	72.28	32.30	19.84	35.23	76.48	19.03	0.00	68.52	8.62	20.62	0.00	71.36	44.10	33.08
UCTransNet [62]	/	64.44	**75.32**	6.27	21.65	39.90	78.96	21.99	0.00	78.99	11.67	28.40	0.00	78.99	45.64	35.80
UM2Former (Ours)	MMT	59.28	73.78	**38.55**	19.02	45.46	**81.75**	20.69	**77.69**	**79.66**	**14.32**	26.00	**53.60**	**79.20**	**61.98**	**48.87**

TABLE XII
SEGMENTATION ACCURACY OF DIFFERENT METHODS IN S8

Methods	Backbone	IoU(%) Score per Category																OA(%)	mF1(%)	mIoU(%)
		Paddy field	Dry farm	Shrubbery	Other forest land	High covered grass land	Medium covered grass land	Low covered grass land	Rival canal	Reservoir	Beach land	Shoal	Urban build-up	Rural settle-ment	Other con-struction land	Bare land	Ocean			
Hyperspectral image classification methods																				
3DCNN [54]	/	1.34	71.35	0.00	0.00	0.00	0.00	0.00	48.89	56.04	19.26	0.00	23.11	26.12	23.26	0.00	86.53	73.21	40.84	33.22
A^{2}S^{2}K-ResNet [55]	/	14.33	76.47	0.00	0.00	0.00	0.00	0.00	37.10	72.32	83.32	22.76	50.33	46.43	29.80	0.00	98.59	79.72	40.84	33.22
Semantic segmentation Convolutional networks																				
FreeNet [39]	/	36.21	77.49	0.00	4.94	26.97	30.43	28.57	53.41	**80.12**	90.48	57.36	55.72	56.86	**51.39**	86.00	99.48	**81.90**	63.72	52.84
UperNet [58]	ResNet-50	21.41	59.17	0.00	7.89	60.50	**47.73**	**99.77**	13.25	34.29	36.99	**64.93**	39.16	40.53	30.78	95.71	87.14	76.84	57.69	46.19
Semantic segmentation Transformer methods																				
TransUNet [60]	ViT-B/16	1.54	76.40	0.00	0.00	0.00	0.00	0.00	0.00	60.10	0.00	0.00	8.87	38.55	19.85	0.00	97.87	74.09	23.04	18.95
U-NetFormer [48]	ResNet-18	29.67	72.79	0.00	0.00	0.00	0.00	13.86	48.77	67.67	69.31	33.92	57.86	62.20	41.90	0.00	97.97	79.93	46.33	37.25
HiViT [61]	HiViT-B	**41.04**	**82.50**	0.00	0.00	0.00	0.00	0.00	50.20	77.83	95.45	49.95	52.42	59.64	46.98	0.00	99.62	81.84	47.15	44.67
TransXNet [63]	TransX-T	36.02	78.88	0.00	0.00	0.00	0.00	20.12	**64.18**	66.18	94.15	47.22	**64.71**	**69.55**	50.14	0.00	94.38	81.60	49.36	43.96
DCSwin [30]	Swin-Tiny	21.02	65.06	0.00	0.00	49.05	36.27	55.99	54.79	59.19	65.37	50.38	46.29	45.22	24.99	0.00	91.27	73.38	53.54	41.55
MANet [59]	ResNet-50	24.40	70.65	0.00	0.00	0.00	0.00	0.00	61.56	78.38	86.95	26.27	54.58	66.32	32.05	0.00	99.39	79.83	44.96	37.53
SegFormer [47]	MIT-B1	27.57	77.18	0.00	0.00	0.00	0.00	0.00	50.75	77.09	90.16	54.52	40.09	47.08	36.62	0.00	99.43	80.73	45.32	37.56
UCTransNet [62]	/	29.33	64.56	0.00	0.00	0.00	0.00	0.85	43.74	73.10	74.48	23.35	53.76	55.70	26.63	83.67	99.02	76.03	48.01	39.62
UM2Former (Ours)	MMT	21.73	75.22	**100.00**	**70.97**	**68.05**	40.50	53.22	57.93	77.14	**95.86**	57.21	54.83	52.24	35.99	**97.67**	**99.88**	81.34	**76.95**	**65.71**

7) Visualizations of Challenging Categories: The segmentation accuracy of UM2Former in *Bare rock*, *Shoal*, *Shrubbery*, and *Bare land* is much higher than other comparison methods. Considering that the ground-truth maps corresponding to the images containing the above categories in the validation set only label one or two categories and have large

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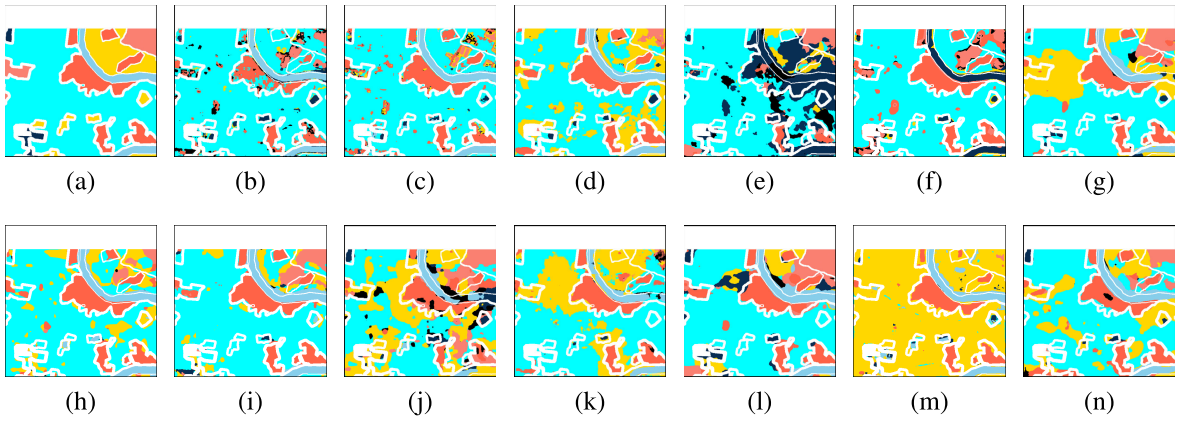


Fig. 11. Visualization results from the S8 test set. (a) Ground truth. (b) 3DCNN (33.22%). (c) A^2S^2K-ResNet (33.22%). (d) FreeNet (52.84%). (e) UperNet (46.19%). (f) TransUNet (18.95%). (g) U-NetFormer (37.25%). (h) HiViT (44.67%). (i) TransXNet (43.96%). (j) DCSwin (41.55%). (k) MANet (37.53%). (l) SegFormer (37.56%). (m) UCTransNet (39.62%). (n) UM2Former (65.71%).

TABLE XIII
SEGMENTATION ACCURACY OF DIFFERENT METHODS IN T5

Methods	Backbone	IoU(%) Score per Category													OA(%)	mF1(%)	mIoU(%)
		Dry farm	High covered grassland	Medium covered grassland	Low covered grassland	Reservoir	Urban build-up	Rural settlement	Other Construction land	Gobi	Saline	Marshland	Bare land	Bare rock			
Hyperspectral image classification methods																	
3DCNN [54]	/	81.56	9.90	69.32	61.22	91.47	23.50	0.00	0.00	56.43	17.38	37.01	0.00	81.38	75.35	49.60	40.70
A^2S^2K-ResNet [55]	/	87.63	36.17	66.25	71.34	89.68	48.15	0.00	0.51	60.54	32.23	70.78	0.00	80.29	79.69	58.94	49.51
Semantic segmentation Convolutional networks																	
FreeNet [39]	/	90.75	43.23	63.70	79.27	**93.86**	58.99	1.52	1.34	65.69	60.66	78.97	50.65	78.32	82.11	69.06	59.15
UperNet [58]	ResNet-50	82.66	18.62	69.75	62.21	9.87	44.86	**28.84**	9.13	51.06	46.46	56.59	22.72	75.56	75.49	57.59	44.49
Semantic segmentation Transformer methods																	
TransUNet [60]	ViT-B/16	81.22	8.62	74.35	69.35	0.00	15.32	0.00	0.00	53.57	**72.19**	0.00	0.00	75.76	78.47	41.47	34.64
U-NetFormer [48]	ResNet-18	85.71	18.34	**82.92**	54.04	76.39	46.28	0.00	0.34	54.54	0.32	0.00	0.00	79.62	73.01	45.74	38.35
HiViT [61]	HiViT-B	86.93	**51.62**	79.64	**80.25**	0.00	47.15	0.00	1.04	66.26	60.30	0.00	0.00	79.89	82.27	49.92	42.66
TransXNet [63]	TransX-T	92.07	38.36	43.83	74.88	92.36	**54.27**	0.00	12.00	60.64	70.83	71.03	0.00	82.93	81.04	62.91	53.32
DCSwin [30]	Swin-Tiny	80.32	21.24	81.64	64.48	46.74	39.54	19.66	2.24	60.17	18.41	13.17	29.72	79.34	76.26	54.91	42.82
MANet [59]	ResNet-50	85.71	47.54	65.45	70.92	60.88	51.87	1.47	1.46	**71.57**	64.52	77.70	0.00	82.15	82.94	62.16	52.40
SegFormer [47]	MIT-B1	**92.33**	36.27	82.39	78.00	26.18	60.27	0.00	21.65	61.33	25.15	77.68	0.00	**86.16**	81.06	59.67	49.80
UCTransNet [62]	/	89.81	46.72	80.36	79.82	0.00	33.08	0.00	0.00	68.59	68.92	70.71	0.00	79.36	82.91	55.41	47.49
UM2Former (Ours)	MMT	91.81	35.49	81.30	72.40	79.89	51.02	10.94	**49.70**	65.98	61.42	**87.61**	**99.52**	82.34	**83.07**	**72.80**	**63.44**

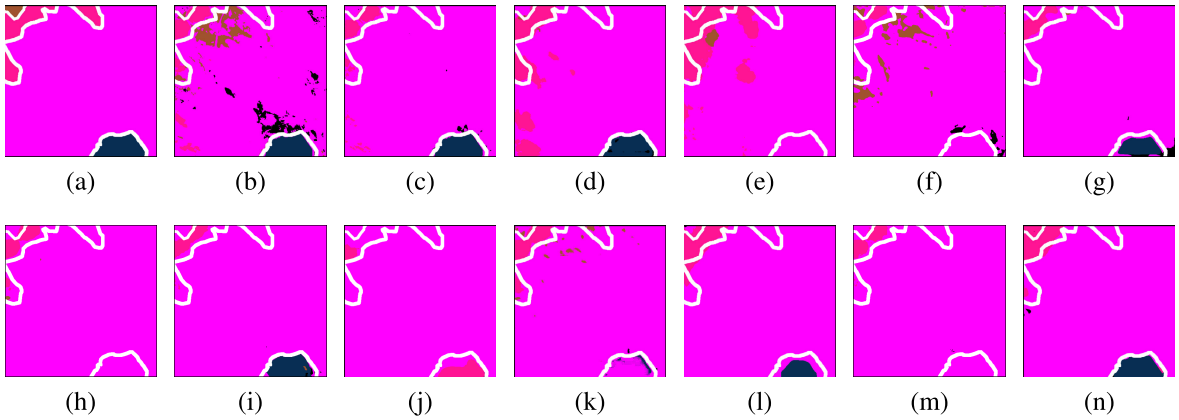


Fig. 12. Visualization results from the T5 test set. (a) Ground truth. (b) 3DCNN (40.70%). (c) A^2S^2K-ResNet (49.51%). (d) FreeNet (59.15%). (e) UperNet (44.49%). (f) TransUNet (34.64%). (g) U-NetFormer (38.35%). (h) HiViT (42.66%). (i) TransXNet (53.32%). (j) DCSwin (42.82%). (k) MANet (52.40%). (l) SegFormer (49.80%). (m) UCTransNet (47.49%). (n) UM2Former (63.44%).

unlabeled areas, we crop the prediction results to visualize the segmentation details of these challenging categories. To analyze whether our method has learned the generalized information and achieved correct segmentation, we visualize the prediction results of FreeNet, MANet, UperNet, and UM2Former. According to Fig. 14, FreeNet can only extract features of *Bare land* and *Shoal*, where most of the *Bare land* is wrong segmentation. Meanwhile, MANet can only identify parts of *Shoal* areas, and both of them identify part of *Shoal* as *River canal*. In addition, UperNet exclusively focuses on processing HSI in the spatial dimension without the comprehensive integration of spectral–spatial information. Consequently, the segmentation results are generally a mess. In comparison with these models, the prediction

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TABLE XIV
SEGMENTATION ACCURACY OF DIFFERENT METHODS IN O21

Methods	Backbone	IoU(%) Score per Category																		OA(%)	mF1(%)	mIoU(%)
		Paddy field	Dry farm	Wood land	Other forest land	High covered grass land	Medium covered grass land	Low covered grass land	River	Lake	Reservoir	Shoal	Urban build-up	Rural settle-ment	Other con-struc-tion land	Sand	Gobi	Bare land	Bare rock			
Hyperspectral image classification methods																						
3DCNN [54]	/	76.85	13.77	0.00	0.00	0.00	19.08	41.99	41.34	36.09	26.82	0.25	34.76	11.82	8.99	0.00	27.63	0.00	36.76	64.79	30.16	20.90
A^{2}S^{2}K-ResNet [55]	/	81.89	35.43	0.00	0.00	0.00	38.01	40.21	60.99	59.29	**28.81**	6.25	57.29	22.95	10.51	0.00	29.40	0.00	**85.67**	71.68	40.47	30.93
Semantic segmentation Convolutional networks																						
FreeNet [39]	/	81.95	42.89	30.13	7.06	**50.44**	44.76	**49.08**	**70.83**	82.97	21.22	**26.26**	67.00	36.01	33.14	19.61	43.46	44.41	57.93	75.14	59.21	44.95
UperNet [58]	ResNet-50	74.60	19.47	38.93	11.88	2.71	11.37	29.50	57.06	81.79	19.25	12.13	43.73	22.54	22.28	**44.20**	21.43	70.23	30.57	61.63	46.84	34.09
Semantic segmentation Transformer methods																						
TransUNet [60]	ViT-B/16	79.44	6.68	0.00	0.00	0.00	0.00	22.79	47.32	0.00	0.00	5.93	47.15	4.96	15.79	0.00	7.87	0.00	22.52	58.66	20.32	14.47
U-NetFormer [48]	ResNet-18	81.57	34.66	0.00	0.00	0.00	18.22	40.73	65.21	0.00	24.47	18.29	68.48	33.96	21.43	0.00	29.71	0.00	62.22	70.23	37.17	27.72
HiViT [61]	HiViT-B	82.83	49.73	0.00	0.00	0.00	45.44	38.63	65.09	0.00	26.01	9.91	58.07	22.39	25.84	0.00	32.80	0.00	69.55	73.12	38.67	29.24
TransXNet [63]	TransX-T	**83.44**	**53.19**	29.47	**13.03**	39.75	45.21	42.06	67.06	87.56	27.53	22.73	50.07	**36.90**	27.83	32.94	49.36	21.54	78.55	75.12	56.61	43.27
DCSwin [30]	Swin-Tiny	78.17	25.60	10.36	11.96	26.76	12.15	28.96	56.62	62.23	24.48	13.22	54.05	24.82	19.92	39.91	36.09	57.17	38.98	66.20	48.40	34.53
MANet [59]	ResNet-50	79.59	33.09	3.28	6.06	8.84	7.96	43.50	69.42	50.27	20.54	20.35	68.66	33.31	17.86	0.43	39.79	8.19	34.31	67.12	41.68	30.30
SegFormer [47]	MIT-B1	69.95	27.38	22.89	9.69	18.95	28.09	30.41	62.41	70.42	19.25	13.42	64.79	27.84	22.97	29.82	29.21	72.36	54.25	63.97	51.28	37.45
UCTransNet [62]	/	82.00	35.68	0.00	0.00	32.44	**46.44**	42.76	66.61	9.94	18.84	19.91	66.33	32.64	33.40	5.79	37.09	14.94	62.22	73.99	45.89	33.76
UM2Former (Ours)	MMT	81.01	38.42	**41.62**	10.69	38.52	43.86	46.48	69.02	**88.83**	23.19	24.68	**71.60**	31.14	**37.81**	33.43	**49.48**	**80.27**	61.26	**75.27**	**62.41**	**48.41**

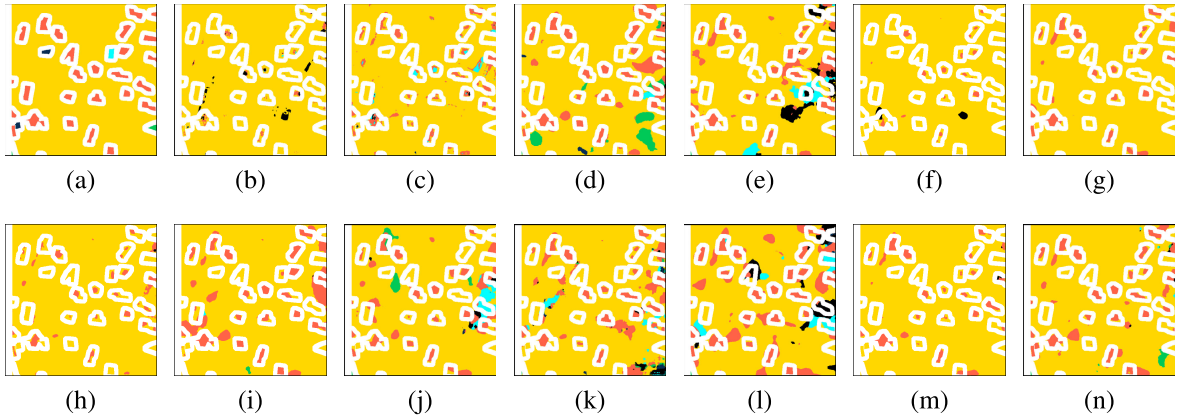


Fig. 13. Visualization results from the O21 test set. (a) Ground truth. (b) 3DCNN (20.90%). (c) A^{2}S^{2}K-ResNet (30.93%). (d) FreeNet (44.95%). (e) UperNet (34.09%). (f) TransUNet (14.47%). (g) U-NetFormer (27.72%). (h) HiViT (29.24%). (i) TransXNet (43.27%). (j) DCSwin (34.53%). (k) MANet (30.30%). (l) SegFormer (37.45%). (m) UCTransNet (33.76%). (n) UM2Former (48.41%).

results of UM2Former demonstrate a favorable distribution and effectively achieve accurate classification across the aforementioned categories, indicating that our method can extract features more precisely. This is because our weight encoder can extract hierarchical discriminative spectral–spatial feature maps and decrease the spectral redundancy, which ensures the sufficient exploitation of both abundant spectral and spatial information in input HSI. Furthermore, benefiting from the powerful global representation capability of MMTB, only UM2Former correctly classified the *Shrubbery* category.

IV. DISCUSSION

In this section, complexity analysis between different methods, as well as attention visualization maps for different models, are discussed. In addition, some considerations about the future work in hyperspectral semantic segmentation are given.

A. Complexity Analysis

The FLOPs (G), parameter size (M), and speed (FPS) of 3DCNN, A^{2}S^{2}K-ResNet, UperNet, TransUNet, U-NetFormer, HiViT, TransXNet, DCSwin, MANet, SegFormer, UCTransNet, and the proposed UM2Former with 512 $\times$ 512 as input in different regions are reported in Table XV. As we can see, traditional hyperspectral classification methods have fewer parameters than other methods and, therefore, are more suitable for small sample classification. Compared with semantic segmentation DL methods, UM2Former has the second-fewest parameters with a relatively fast inference speed. Although UM2Former performs a slower inference speed than Transformer-based semantic segmentation models like U-NetFormer, DCSwin, and SegFormer, the model parameter of UM2Former is only 57% of U-NetFormer, 14% of DCSwin, and 49% of SegFormer. Furthermore, in comparison with the other U-shaped Transformer models such as HiViT, TransXNet, TransUNet, and UCTransNet, the proposed UM2Former achieves a competitive inference speed while possessing over 70% fewer model parameters. Several reasons are given to explain the superiority of UM2Former in terms of complexity, parameter size, and speed. First, UM2Former employs a lateral connection to control the dimension of the weighted feature maps embedded into the decoder, which is beneficial as the redundant and unnecessary channel information will greatly reduce the inference speed. Second, the core module of MMTB adopts average pooling operation to retent spatial information, greatly reducing model parameters compared with traditional MHSA. Third,

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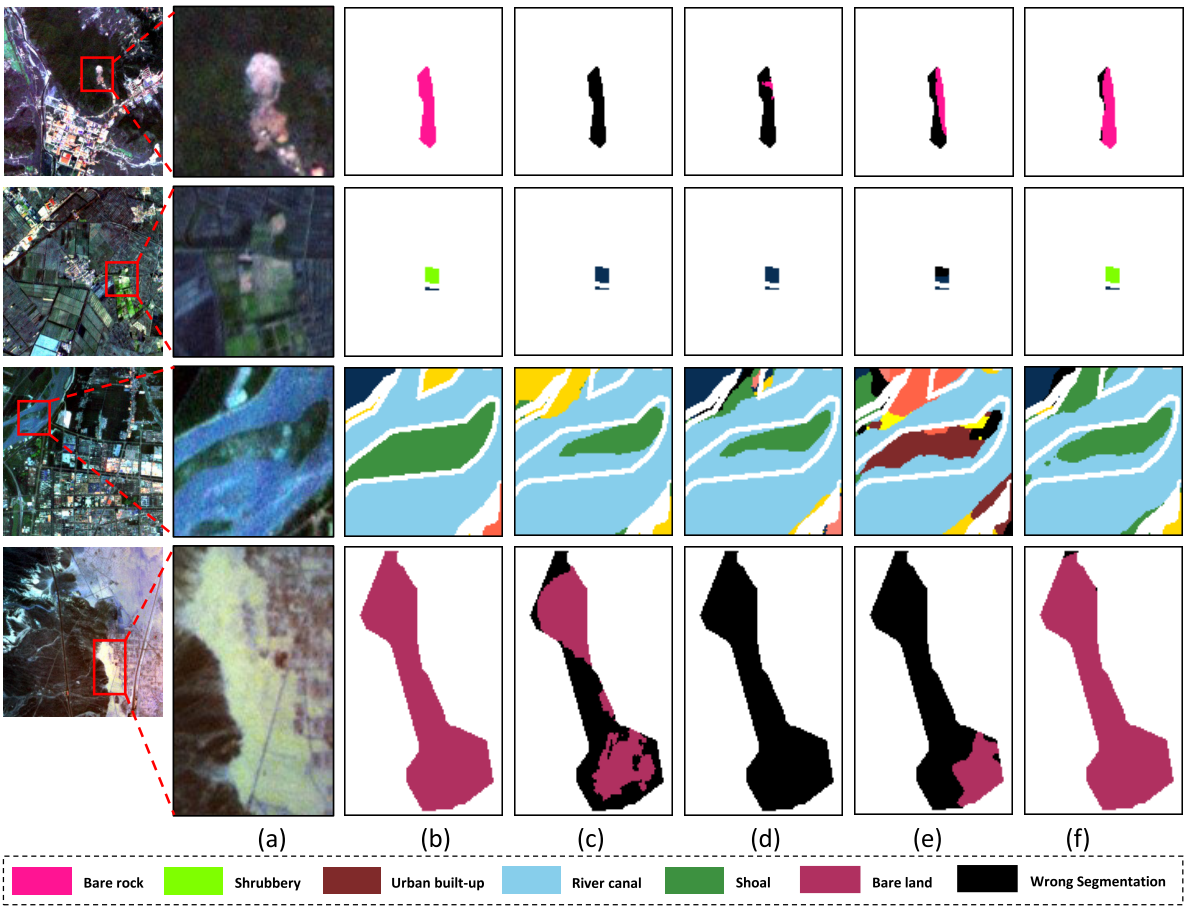


Fig. 14. Visualization results of challenging categories obtained by different models from T1, S8, O21, and T5 validation sets. (Top row) T1_0105. (Second row) S8_0035. (Third row) O21_0108. (Bottom row) T5_0106. (a) RGB. (b) Label. (c) FreeNet. (d) MANet. (e) UperNet. (f) UM2Former.

TABLE XV
COMPLEXITY, PARAMETER SIZE, AND SPEED OF DIFFERENT MODELS

Methods	Metrics						
	FLOPs(G)	Parameter(M)	Speed(FPS)				
			T1	S3	S8	T5	O21
Hyperspectral image classification methods							
3DCNN [54]	1814.51	0.35	3.53	7.17	3.60	3.59	3.24
A^2S^2K-ResNet [55]	93.41	0.11	2.93	3.35	2.26	2.25	2.73
Semantic segmentation Convolutional networks							
FreeNet [39]	85.07	15.14	8.53	54.33	32.20	17.59	9.07
UperNet [58]	49.98	27.53	16.55	47.36	32.44	20.85	21.00
Semantic segmentation Transformer methods							
TransUNet [60]	101.15	36.56	7.90	34.88	21.78	13.71	5.95
U-NetFormer [48]	17.72	11.77	30.06	144.19	74.07	69.47	38.21
HiViT [61]	58.27	34.42	3.71	33.71	5.52	20.80	11.05
TransXNet [63]	18.32	19.28	15.64	15.74	15.70	15.87	15.69
DCSwin [30]	47.73	45.65	11.74	71.48	42.07	22.68	11.96
MANet [59]	85.08	35.96	19.89	52.34	33.05	19.25	20.30
SegFormer [47]	14.83	13.77	12.43	78.45	55.03	28.86	14.99
UCTransNet [62]	143.99	34.66	4.46	13.13	8.34	4.32	3.98
UM2Former (Ours)	39.52	6.77	12.17	56.84	44.49	23.18	15.79

the segmentation head, LFSH, is implemented by several linear layers, few learnable parameters, and two convolution layers, which fuses multilevel semantic information effectively without complex and computationally demanding modules.

Above all, UM2Former achieves great performance of large-scale HSI semantic segmentation without consuming excessive computing resources and occupying excessive storage resources.

B. Feature Visualization

The advantage of the SFRA mechanism in the proposed MMTB is that it can retent the average spatial characteristics of different locations and then model the global dependency of each pixel in the input feature maps with these characteristics. To demonstrate this, we visualize the self-attention feature maps from layers of different self-attention mechanisms in

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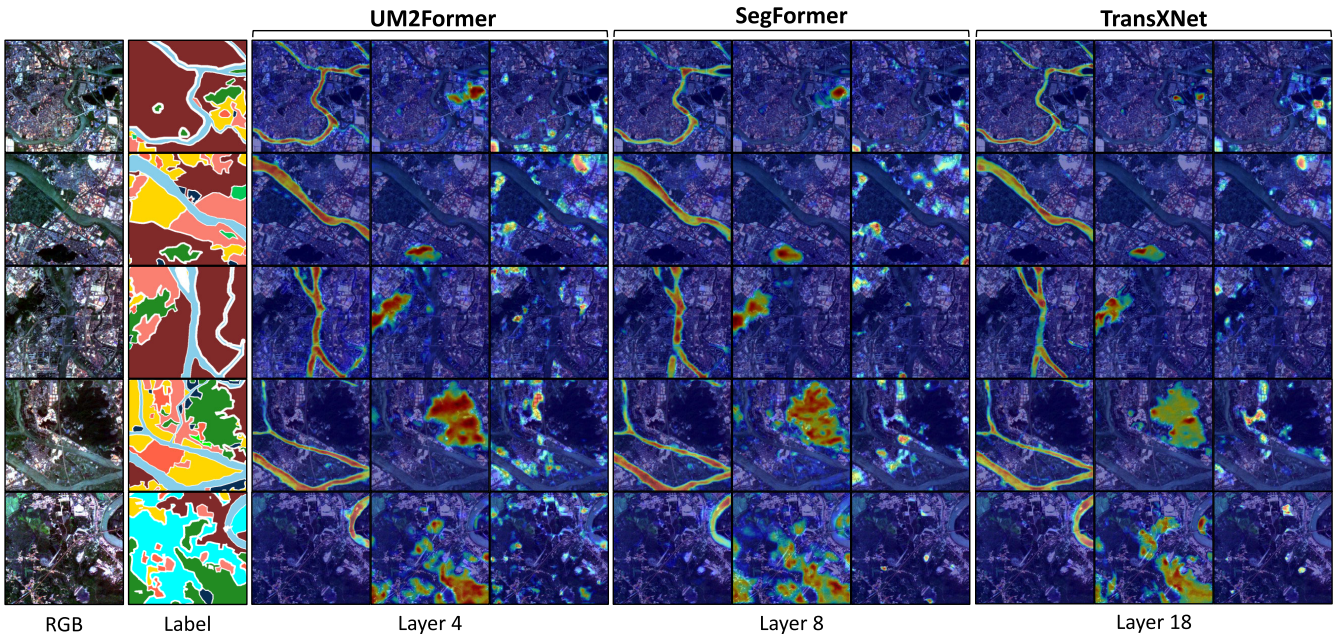


Fig. 15. Attention maps of trained UM2Former, SegFormer, and TransXNet on images from WHU-OHS. We select the fourth attention maps layer for UM2Former, the eighth attention maps layer for SerFormer, and the eighteenth attention maps for TransXNet.

TABLE XVI
SEGMENTATION ACCURACY OF SELECTED METHODS ON ALL 42 CITIES

Class	HSI Classification Methods		Semantic Segmentation CNN Networks		Semantic Segmentation Transformer Networks			
	3DCNN [54]	A^2S^2K-ResNet [55]	UperNet [58]	FreeNet [39]	TransUNet [60]	SegFormer [47]	DCSwin [30]	UM2Former
Paddy field	45.61	61.96	62.61	72.16	**72.98**	64.87	71.84	72.37
Dry farm	41.38	63.64	62.89	**74.69**	70.29	67.28	67.42	72.43
Woodland	45.69	51.68	55.10	63.24	64.78	56.92	64.56	**67.04**
Shrubbery	13.96	20.32	28.68	37.14	36.78	36.89	**42.73**	40.38
Sparse woodland	0.00	0.00	13.70	13.28	6.96	17.46	9.63	**17.87**
Other forestland	0.00	0.00	2.98	**18.60**	0.00	12.37	9.74	11.33
High-covered Grassland	33.27	39.28	41.86	**57.12**	46.97	50.74	53.29	54.66
Medium-covered Grassland	10.67	23.68	20.10	28.46	28.73	26.93	28.14	**33.20**
Low-covered Grassland	26.37	31.12	32.91	**47.25**	31.46	42.14	42.74	41.51
River/canal	43.96	36.32	57.82	61.70	63.28	54.20	**66.34**	63.85
Lake	78.77	71.89	75.06	87.28	81.38	**89.72**	89.67	85.66
Reservoir/pond	24.38	27.87	41.25	56.48	52.71	44.79	48.65	**57.13**
Beach land	0.00	0.00	52.93	48.39	31.46	41.63	**64.51**	62.40
Shoal	25.69	23.73	43.90	54.23	53.33	46.49	55.55	**57.69**
Urban build-up	27.49	68.14	71.71	**79.64**	77.28	72.15	76.32	77.62
Rural settlement	10.69	18.32	46.48	54.12	51.96	40.12	48.83	**54.88**
Other Construction land	6.48	6.68	29.94	39.46	43.27	36.27	32.99	**39.56**
Sand	48.63	61.23	64.99	**78.77**	29.27	73.59	74.32	69.90
Gobi	43.77	57.28	57.91	68.91	68.39	59.47	**76.12**	64.13
Saline/Alkali soil	25.72	7.28	35.10	43.95	27.23	32.64	**52.73**	51.62
Marshland	13.21	18.29	30.27	53.28	41.76	36.17	**46.34**	42.95
Bare land	0.00	0.00	40.35	29.43	0.00	32.91	17.46	**41.13**
Bare rock	39.24	52.32	62.75	70.14	70.86	70.14	73.27	**73.97**
Ocean	80.18	86.79	84.68	98.36	94.32	93.27	**97.87**	97.06
OA	55.96	67.98	70.96	78.18	75.36	73.18	77.68	**78.43**
mIoU	28.55	34.49	47.23	55.38	47.71	49.96	54.67	**56.18**

Fig. 15. As we can see, the self-attention maps of UM2Former contain clearer segmentation masks in contrast to those of SegFormer and TransXNet, and they exhibit higher contrast and fewer redundant features, indicating that our method can highlight and extract feature more precisely.

C. Comparison on All Cities

In this section, we validate UM2Former across all 42 cities, with 4821 subimages to train and 515 subimages to test, respectively. As shown in Table XVI, the classwise performance is reported via the average of the IoU score based on the test datasets. In this table, the best results are highlighted. First, we focus on representative DL methods under different frameworks. Compared to the single-city experiments, the performance of Transformer-based networks improves significantly, as the increase in training samples enables the model to learn better complex land cover class information and long-range dependencies between pixels. In addition, patch-based HSI classification networks have achieved great success in single-scene classification tasks under few-short learning conditions, thanks to their strong

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ability to mine spectral–spatial features. However, because of their relatively fewer parameters, they exhibit limited generalization performance, making them unsuitable for HSI semantic segmentation tasks. Our method combines the advantages of both, achieving promising results in large-scale HSI semantic segmentation tasks. We then compare the classwise characteristics. The ocean and lake categories have desirable IoU scores, mainly due to their distinctive spectral characteristics. Under the combined influence of spectral and spatial information, for instance, bare land may exhibit a more uniform soil tone, while bare rock typically displays more distinct surface textures. Our method achieves the best IoU scores for these categories. Furthermore, with the assistance of spatial information, UM2Former and FreeNet are able to effectively distinguish between *Urban build-up* and *Rural settlement*.

D. Future Perspective

Experiments are conducted in five study areas of Jilin, Guangzhou, Tianjin, Kelamayi, and Wuzhong. Compared with the state-of-the-art semantic segmentation algorithms, the proposed method achieves higher accuracies. However, there exist limitations in this study. Limited by the scarcity of pretrained networks in HSI, methods that are effective for RGB image segmentation lack generalization capability when applied to HSI task. It is worth mentioning that there are many pretrained network models in high-resolution RS images. In the future, we will work toward the large-scale pretraining backbone network of HSI to extract generalizable features for single-scene HSI small-sample classification tasks.

V. Conclusion

In this article, we propose an UM2Former for HSI semantic segmentation. First, weight encoder is designed to extract hierarchical discriminative spectral–spatial feature maps over the entire input HSI and decrease the spectral redundancy. Second, our lateral connection preserves the most effective feature maps and provides more precise locations of features for the decoder. Third, SFRA module is proposed to retent the spatial average characteristics of different locations and then model the global dependency of each pixel in the input feature maps. Then, WF operation is utilized during the upsampling process to obtain a generalized feature representation. Finally, LFSH with multilinear layers and semantic fusion module is used to fuse multilevel semantic features and achieve segmentation. UM2Former is a model specially designed for HSI semantic segmentation. Experimental and visualization results conducted on WHU-OHS validated that our method can achieve better segmentation performance compared with state-of-the-art DL-based semantic segmentation methods.

Although we have obtained encouraged experimental results, we will investigate the generalization ability of the proposed UM2Former in future by transfer learning methods such as domain adaptation and domain generalization.

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Aijun Xu received the B.S. degree from the School of Marine Technology and Geomatics, Jiangsu Ocean University, Lianyungang, China, in 2023. He is currently pursuing the M.E. degree in mapping and remote sensing engineering with Hohai University, Nanjing, China.

His research interests include hyperspectral image semantic segmentation and computer vision.



Zhaohui Xue (Member, IEEE) received the B.S. degree in geomatics engineering from Shandong Agricultural University, Tai'an, China, in 2009, the M.E. degree in remote sensing from China University of Mining and Technology, Beijing, China, in 2012, and the Ph.D. degree in cartography and geographic information system from Nanjing University, Nanjing, China, in 2015.

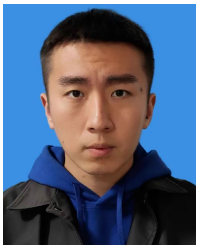
He is currently a Professor (Ph.D. Supervisor) at the College of Geography and Remote Sensing, Hohai University, Nanjing. He has authored more than 70 scientific papers including more than 40 Science Citation Index (SCI) papers. His research interests include hyperspectral image classification, time-series image analysis, pattern recognition, and machine learning.

Dr. Xue was a recipient of the National Scholarship for Doctoral Graduate Students granted by the Ministry of Education of China in 2014. He was awarded the Best Reviewer for the IEEE Geoscience and Remote Sensing Society. He is an Editorial Board Member of NATIONAL REMOTE SENSING BULLETIN from 2020 to 2024. He has been a reviewer for more than ten famous remote sensing journals including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and *ISPRS Journal of Photogrammetry and Remote Sensing*.



Ziyu Li received the B.S. degree in remote sensing science and technology from Nanjing University of Information Science and Technology, Nanjing, China, in 2022. He is currently pursuing the M.E. degree in surveying and mapping with Hohai University, Nanjing.

His research interests include hyperspectral image classification and cross-view remote sensing image geolocalization.



Shun Cheng received the B.S. degree in remote sensing science and technology from Nanjing Tech University, Nanjing, China, in 2023. He is currently pursuing the M.E. degree in mapping and remote sensing engineering with Hohai University, Nanjing.

His research interests include hyperspectral image classification and machine learning.



Hongjun Su (Senior Member, IEEE) received the Ph.D. degree in cartography and geography information system from the Key Laboratory of Virtual Geographic Environment (Ministry of Education), Nanjing Normal University, Nanjing, China, in 2011.

He is currently a Professor at the College of Geography and Remote Sensing, Hohai University, Nanjing. His main research interests include hyperspectral remote sensing dimensionality reduction, classification, and spectral unmixing.

Dr. Su received the 2016 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society. He is an Associate Editor of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.



Junshi Xia (Senior Member, IEEE) received the B.S. degree in geographic information systems and the Ph.D. degree in photogrammetry and remote sensing from China University of Mining and Technology, Xuzhou, China, in 2008 and 2013, respectively, and the Ph.D. degree in image processing from Grenoble Images Speech Signals and Automatics Laboratory, Grenoble Institute of Technology, Grenoble, France, in 2014.

From 2014 to 2015, he was a Visiting Scientist with the Department of Geographic Information Sciences, Nanjing University, Nanjing, China. From 2015 to 2016, he was a Post-Doctoral Research Fellow with the University of Bordeaux, Bordeaux, France. From 2016 to 2018, he was the Japan Society for the Promotion of Science (JSPS) Postdoctoral Overseas Research Fellow with The University of Tokyo, Kashiwa, Japan. Since 2018, he has been with the RIKEN Center for Advanced Intelligence Project (AIP), Tokyo, Japan, where he is currently a Senior Research Scientist. His research interests include multiple classifier systems in remote sensing, hyperspectral remote sensing image processing, and deep learning in remote sensing applications.

Dr. Xia was a recipient of the First Place Prize in the IEEE Geoscience and Remote Sensing Society Data Fusion Contest organized by the Image Analysis and Data Fusion Technical Committee in 2017. Since 2019, he has been an Associate Editor of IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL), *Remote Sensing*, and *Frontiers in Remote Sensing*, and a Guest Editor of *Remote Sensing* and IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS).

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